

1st Part Problem Set for Applied Abstract Algebra

Quivers: Representations and Morphisms

1. Let $M, M' \in \text{Rep } Q$. Show that the set of morphisms $\text{Hom}(M, M')$ is a k -vector space.

Proof. Let $M = (M_i, \phi_\alpha)$ and $M' = (M'_i, \psi_\alpha)$. $\forall f, g \in \text{Hom}(M, M')$ since f, g are morphisms, for any arrow $\alpha : i \rightarrow j$, we have $\psi_\alpha f_i = f_j \phi_\alpha$ and $\psi_\alpha g_i = g_j \phi_\alpha$.

(a) **Addition:**

$$\psi_\alpha(f + g)_i = \psi_\alpha(f_i + g_i) = \psi_\alpha f_i + \psi_\alpha g_i = f_j \phi_\alpha + g_j \phi_\alpha = (f_j + g_j) \phi_\alpha = (f + g)_j \phi_\alpha.$$

Thus $f + g \in \text{Hom}(M, M')$.

(b) **Scalar Multiplication:**

$$\psi_\alpha(\lambda f)_i = \psi_\alpha(\lambda f_i) = \lambda(\psi_\alpha f_i) = \lambda(f_j \phi_\alpha) = (\lambda f_j) \phi_\alpha = (\lambda f)_j \phi_\alpha.$$

Thus $\lambda f \in \text{Hom}(M, M')$.

The zero morphism $0 = (0)_{i \in Q_0}$ clearly satisfies the commutativity condition. The vector space axioms follow from the structure of linear maps between vector spaces. $\square$

2. Let Q be the quiver

$$1 \longrightarrow 2 \longleftarrow 3$$

and consider the representations

$$M: \quad k \xrightarrow{\begin{pmatrix} 1 \\ 0 \end{pmatrix}} k^2 \xleftarrow{\begin{pmatrix} 0 \\ 1 \end{pmatrix}} k \quad \quad M': \quad k \xrightarrow{\begin{pmatrix} 1 \\ 0 \end{pmatrix}} k^2 \xleftarrow{\begin{pmatrix} 1 \\ 0 \end{pmatrix}} k$$

(a) Show that M and M' are not indecomposable.

Proof. We exhibit a nontrivial direct sum decomposition for each representation.

Decomposition of M . Define subrepresentations

$$N_1: \quad k \xrightarrow{\begin{pmatrix} 1 \\ 0 \end{pmatrix}} \text{span}(e_1) \xleftarrow{0} 0, \quad \quad N_2: \quad 0 \xrightarrow{0} \text{span}(e_2) \xleftarrow{\begin{pmatrix} 0 \\ 1 \end{pmatrix}} k.$$

At each vertex we have $M(1) = k = N_1(1) \oplus N_2(1)$, $M(2) = k^2 = \text{span}(e_1) \oplus \text{span}(e_2) = N_1(2) \oplus N_2(2)$, and $M(3) = k = N_1(3) \oplus N_2(3)$. The linear maps of M respect this decomposition: the map $\begin{pmatrix} 1 \\ 0 \end{pmatrix}: k \rightarrow k^2$ lands in $\text{span}(e_1) = N_1(2)$, and

$\begin{pmatrix} 0 \\ 1 \end{pmatrix}: k \rightarrow k^2$ lands in $\text{span}(e_2) = N_2(2)$. Hence $M = N_1 \oplus N_2 \cong P(1) \oplus P(3)$, so M is not indecomposable.

Decomposition of M' . Define subrepresentations

$$N'_1: \quad k \xrightarrow{1} \text{span}(e_1) \xleftarrow{1} k, \quad N'_2: \quad 0 \xrightarrow{0} \text{span}(e_2) \xleftarrow{0} 0.$$

At each vertex: $M'(1) = k = N'_1(1) \oplus N'_2(1)$, $M'(2) = k^2 = \text{span}(e_1) \oplus \text{span}(e_2) = N'_1(2) \oplus N'_2(2)$, and $M'(3) = k = N'_1(3) \oplus N'_2(3)$. Both arrows $\begin{pmatrix} 1 \\ 0 \end{pmatrix}: M'(1) \rightarrow M'(2)$ and $\begin{pmatrix} 1 \\ 0 \end{pmatrix}: M'(3) \rightarrow M'(2)$ land in $\text{span}(e_1) = N'_1(2)$, so the maps respect the decomposition. Hence $M' = N'_1 \oplus N'_2 \cong I(2) \oplus S(2)$, so M' is not indecomposable. $\square$

(b) Show that M and M' are not isomorphic.

Proof. Suppose for contradiction that $f: M \rightarrow M'$ is an isomorphism, so $f = (f_1, f_2, f_3)$ with f_i invertible. The commutativity conditions give:

$$f_2\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}f_1, \quad f_2\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}f_3.$$

Let $f_1 = (a) \in k^*$, $f_3 = (c) \in k^*$, and $f_2 = \begin{pmatrix} p & q \\ r & s \end{pmatrix} \in \text{GL}_2(k)$.

From the first equation: $f_2\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} p \\ r \end{pmatrix} = \begin{pmatrix} a \\ 0 \end{pmatrix}$, so $p = a$ and $r = 0$.

From the second equation: $f_2\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} q \\ s \end{pmatrix} = \begin{pmatrix} c \\ 0 \end{pmatrix}$, so $q = c$ and $s = 0$.

Then $f_2 = \begin{pmatrix} a & c \\ 0 & 0 \end{pmatrix}$, which has $\det(f_2) = 0$. This contradicts $f_2 \in \text{GL}_2(k)$.

Therefore $M \not\cong M'$. $\square$

3. Let Q be the quiver

$$1 \longrightarrow 2 \longleftarrow 3$$

and let M be the representation

$$M: \quad k^2 \xrightarrow{\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}} k^3 \xleftarrow{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}} k.$$

(a) Write M as a direct sum of the indecomposable representations listed below:

$$\begin{array}{lll} S(1): k \rightarrow 0 \leftarrow 0 & S(2): 0 \rightarrow k \leftarrow 0 & S(3): 0 \rightarrow 0 \leftarrow k \\ P(1): k \xrightarrow{1} k \leftarrow 0 & I(2): k \xrightarrow{1} k \xleftarrow{1} k & P(3): 0 \rightarrow k \xleftarrow{1} k \end{array}$$

Proof. We claim that

$$M \cong I(2) \oplus P(1) \oplus S(2).$$

The dimension vectors of the summands are

$$\underline{\dim} I(2) = (1, 1, 1), \quad \underline{\dim} P(1) = (1, 1, 0), \quad \underline{\dim} S(2) = (0, 1, 0),$$

and their sum is

$$(1, 1, 1) + (1, 1, 0) + (0, 1, 0) = (2, 3, 1) = \underline{\dim} M.$$

We now verify that the maps of $I(2) \oplus P(1) \oplus S(2)$ agree with those of M . Order the components at each vertex as:

$$\text{Vertex 1: } I(2)(1) \oplus P(1)(1) = k \oplus k = k^2,$$

$$\text{Vertex 2: } I(2)(2) \oplus P(1)(2) \oplus S(2)(2) = k \oplus k \oplus k = k^3,$$

$$\text{Vertex 3: } I(2)(3) = k.$$

Map at arrow $1 \rightarrow 2$. In the direct sum, the map $k^2 \rightarrow k^3$ sends

$$(a, b) \mapsto (I(2)(\alpha)(a), P(1)(\alpha)(b), 0) = (a, b, 0),$$

since both $I(2)$ and $P(1)$ have identity maps at this arrow, and $S(2)(1) = 0$ contributes nothing. In matrix form this gives

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix},$$

which agrees with the map of M .

Map at arrow $3 \rightarrow 2$. In the direct sum, the map $k \rightarrow k^3$ sends

$$c \mapsto (I(2)(\beta)(c), 0, 0) = (c, 0, 0),$$

since $I(2)$ has identity map at this arrow, and $P(1)(3) = S(2)(3) = 0$ contribute nothing. In matrix form this gives

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix},$$

which agrees with the map of M . Therefore $M \cong I(2) \oplus P(1) \oplus S(2)$. $\square$

(b) Show that there is a non-split short exact sequence

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0.$$

Proof. Consider the representations

$$S(2) = (0, k, 0), \quad P(1) = (k, k, 0) \text{ with map } \text{id}_k, \quad S(1) = (k, 0, 0).$$

Define morphisms $\iota: S(2) \rightarrow P(1)$ and $p: P(1) \rightarrow S(1)$ by

$$\iota = (\iota_1, \iota_2, \iota_3) = (0, \text{id}_k, 0), \quad p = (p_1, p_2, p_3) = (\text{id}_k, 0, 0).$$

Morphism verification. The only nontrivial arrow is $\alpha: 1 \rightarrow 2$ with $P(1)(\alpha) = \text{id}_k$. For ι , the commutativity diagram at α is

$$\begin{array}{ccc} S(2)(1) = 0 & \xrightarrow{0} & S(2)(2) = k \\ \iota_1=0 \downarrow & & \downarrow \iota_2=\text{id}_k \\ P(1)(1) = k & \xrightarrow{\text{id}_k} & P(1)(2) = k \end{array}$$

Both compositions give 0, so the diagram commutes. For p :

$$\begin{array}{ccc} P(1)(1) = k & \xrightarrow{\text{id}_k} & P(1)(2) = k \\ p_1=\text{id}_k \downarrow & & \downarrow p_2=0 \\ S(1)(1) = k & \xrightarrow{0} & S(1)(2) = 0 \end{array}$$

Both compositions give 0, so the diagram commutes.

Exactness.

ι is injective: the only nontrivial component is $\iota_2 = \text{id}_k$, which is injective.

p is surjective: $p_1 = \text{id}_k$ is surjective, and $p_2: k \rightarrow 0$, $p_3: 0 \rightarrow 0$ are trivially surjective.

Exactness at $P(1)$: we verify $\ker p = \text{im } \iota$ at each vertex.

- Vertex 1: $\ker(\text{id}_k) = 0 = \text{im}(0)$.
- Vertex 2: $\ker(0: k \rightarrow 0) = k = \text{im}(\text{id}_k)$.
- Vertex 3: both are 0.

Hence

$$0 \longrightarrow S(2) \xrightarrow{\iota} P(1) \xrightarrow{p} S(1) \longrightarrow 0$$

is a short exact sequence.

Non-split. If the sequence splits, then

$$P(1) \cong S(1) \oplus S(2).$$

However, the direct sum $S(1) \oplus S(2) = (k, k, 0)$ has map $0 \oplus 0 = 0$ at arrow $1 \rightarrow 2$, while $P(1)$ has map $\text{id}_k \neq 0$. This is a contradiction, so the sequence does not split. $\square$

4. Find all indecomposable representations up to isomorphism of the quiver $1 \rightarrow 2$.

Proof. The indecomposable representations of $Q: 1 \rightarrow 2$ are, up to isomorphism:

$$S(1): k \xrightarrow{0} 0, \quad S(2): 0 \xrightarrow{0} k, \quad P(1): k \xrightarrow{\text{id}} k.$$

Step 1: Indecomposability.

$S(1)$ and $S(2)$ have dimension vectors $(1, 0)$ and $(0, 1)$ respectively. Since no component can be split further, these are clearly indecomposable.

For $P(1)$ with dimension vector $(1, 1)$, suppose

$$P(1) = N_1 \oplus N_2$$

is a nontrivial decomposition. Then we must have

$$\underline{\dim} N_1 = (1, 0), \quad \underline{\dim} N_2 = (0, 1)$$

(or vice versa). In this decomposition, the map $\phi: P(1)(1) = k \rightarrow P(1)(2) = k$ would decompose as

$$\phi = \phi|_{N_1} \oplus \phi|_{N_2} = 0 \oplus 0 = 0,$$

since $\phi|_{N_1}: k \rightarrow 0$ and $\phi|_{N_2}: 0 \rightarrow k$ are both the zero map. This contradicts $\phi = \text{id}_k \neq 0$. Hence $P(1)$ is indecomposable.

Step 2: Completeness.

Let

$$M = (V_1, V_2, \phi)$$

be any representation with $\phi: V_1 \rightarrow V_2$. Set

$$n_1 = \dim V_1, \quad n_2 = \dim V_2, \quad r = \text{rank}(\phi).$$

Choose a basis $\{e_1, \dots, e_{n_1}\}$ of V_1 such that $\{e_1, \dots, e_r\}$ maps to linearly independent vectors in V_2 and $\{e_{r+1}, \dots, e_{n_1}\}$ spans $\ker \phi$. Let

$$f_i = \phi(e_i) \quad \text{for } 1 \leq i \leq r,$$

and extend $\{f_1, \dots, f_r\}$ to a basis $\{f_1, \dots, f_{n_2}\}$ of V_2 . In these bases,

$$\phi = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}.$$

Then M decomposes as:

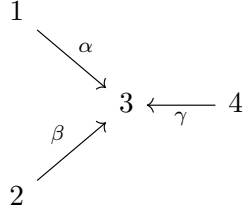
- r copies of $P(1)$: the i -th copy is spanned by e_i at vertex 1 and f_i at vertex 2, with $\phi(e_i) = f_i$, for $1 \leq i \leq r$.
- $(n_1 - r)$ copies of $S(1)$: spanned by e_{r+j} at vertex 1, with $\phi(e_{r+j}) = 0$.
- $(n_2 - r)$ copies of $S(2)$: spanned by f_{r+l} at vertex 2.

Hence

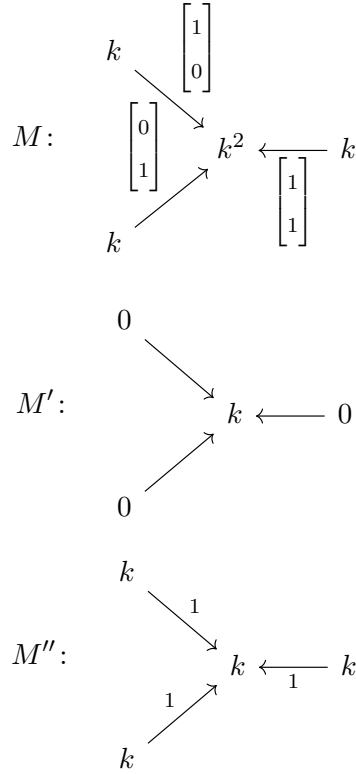
$$M \cong P(1)^{\oplus r} \oplus S(1)^{\oplus (n_1 - r)} \oplus S(2)^{\oplus (n_2 - r)}.$$

Since every representation decomposes into copies of $S(1)$, $S(2)$, and $P(1)$, these are all the indecomposable representations up to isomorphism. $\square$

5. Let Q be the quiver



and consider the following representations



(a) Show that

$$\begin{aligned}
 \operatorname{Hom}(M, M') &= 0, & \operatorname{Hom}(M', M) &\cong k^2 \\
 \operatorname{Hom}(M, M'') &\cong k^2, & \operatorname{Hom}(M'', M) &= 0.
 \end{aligned}$$

Proof. The quiver Q has arrows $\alpha: 1 \rightarrow 3$, $\beta: 2 \rightarrow 3$, $\gamma: 4 \rightarrow 3$. A morphism $f: A \rightarrow B$ consists of linear maps (f_1, f_2, f_3, f_4) satisfying

$$f_3 \circ A(\delta) = B(\delta) \circ f_s$$

for each arrow $\delta: s \rightarrow 3$.

$\operatorname{Hom}(M, M') = 0$.

A morphism $f: M \rightarrow M'$ has components

$$f_1: k \rightarrow 0, \quad f_2: k \rightarrow 0, \quad f_3: k^2 \rightarrow k, \quad f_4: k \rightarrow 0.$$

Since the codomain is 0, we have $f_1 = f_2 = f_4 = 0$. The commutativity condition at α gives

$$\begin{array}{ccc} M(1) = k & \xrightarrow{\begin{pmatrix} 1 \\ 0 \end{pmatrix}} & M(3) = k^2 \\ f_1=0 \downarrow & & \downarrow f_3 \\ M'(1) = 0 & \xrightarrow{0} & M'(3) = k \end{array} \implies f_3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 0.$$

Similarly, the commutativity condition at β gives

$$f_3 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0.$$

Since $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ form a basis of k^2 , we conclude $f_3 = 0$. Therefore $f = 0$.

$\text{Hom}(M', M) \cong k^2$.

A morphism $f: M' \rightarrow M$ has components

$$f_1: 0 \rightarrow k, \quad f_2: 0 \rightarrow k, \quad f_3: k \rightarrow k^2, \quad f_4: 0 \rightarrow k.$$

Since the domains of f_1, f_2, f_4 are 0, these maps are all zero. For each arrow $\delta: s \rightarrow 3$, the commutativity condition becomes

$$f_3 \circ M'(\delta) = M(\delta) \circ f_s.$$

Since $M'(s) = 0$ for $s = 1, 2, 4$, both $M'(\delta)$ and f_s are the zero map, so the condition is automatically satisfied. Hence f is completely determined by the arbitrary linear map

$$f_3 \in \text{Hom}_k(k, k^2) \cong k^2.$$

$\text{Hom}(M, M'') \cong k^2$.

A morphism $f: M \rightarrow M''$ has components

$$f_1 = a: k \rightarrow k, \quad f_2 = b: k \rightarrow k, \quad f_3 = \begin{pmatrix} c & d \end{pmatrix}: k^2 \rightarrow k, \quad f_4 = e: k \rightarrow k,$$

where $a, b, c, d, e \in k$. The commutativity conditions give:

At $\alpha: 1 \rightarrow 3$:

$$f_3 \circ M(\alpha) = M''(\alpha) \circ f_1 \implies \begin{pmatrix} c & d \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1 \cdot a \implies c = a.$$

At $\beta: 2 \rightarrow 3$:

$$f_3 \circ M(\beta) = M''(\beta) \circ f_2 \implies \begin{pmatrix} c & d \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 1 \cdot b \implies d = b.$$

At $\gamma: 4 \rightarrow 3$:

$$f_3 \circ M(\gamma) = M''(\gamma) \circ f_4 \implies \begin{pmatrix} c & d \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1 \cdot e \implies e = a + b.$$

Hence every morphism f is determined by the pair $(a, b) \in k^2$, with $c = a$, $d = b$, $e = a + b$. Therefore

$$\text{Hom}(M, M'') \cong k^2.$$

$\text{Hom}(M'', M) = 0$.

A morphism $f: M'' \rightarrow M$ has components

$$f_1 = a: k \rightarrow k, \quad f_2 = b: k \rightarrow k, \quad f_3 = \begin{pmatrix} c \\ d \end{pmatrix}: k \rightarrow k^2, \quad f_4 = e: k \rightarrow k.$$

At $\alpha: 1 \rightarrow 3$:

$$M(\alpha) \circ f_1 = f_3 \circ M''(\alpha) \implies \begin{pmatrix} 1 \\ 0 \end{pmatrix} a = \begin{pmatrix} c \\ d \end{pmatrix} \cdot 1 \implies c = a, d = 0.$$

At $\beta: 2 \rightarrow 3$:

$$M(\beta) \circ f_2 = f_3 \circ M''(\beta) \implies \begin{pmatrix} 0 \\ 1 \end{pmatrix} b = \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a \\ 0 \end{pmatrix} \implies a = 0, b = 0.$$

At $\gamma: 4 \rightarrow 3$:

$$M(\gamma) \circ f_4 = f_3 \circ M''(\gamma) \implies \begin{pmatrix} 1 \\ 1 \end{pmatrix} e = \begin{pmatrix} a \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies e = 0.$$

All components vanish, so $\text{Hom}(M'', M) = 0$. □

(b) Show that M is not isomorphic to $M' \oplus M''$.

Proof. Suppose for contradiction that

$$M \cong M' \oplus M''.$$

Then by the additivity of Hom ,

$$\text{Hom}(M, M') \cong \text{Hom}(M' \oplus M'', M') \cong \text{Hom}(M', M') \oplus \text{Hom}(M'', M').$$

Since $M' \neq 0$, the identity morphism $\text{id}_{M'}$ is a nonzero element of $\text{Hom}(M', M')$. Hence

$$\text{Hom}(M', M') \neq 0,$$

which implies

$$\text{Hom}(M, M') \neq 0.$$

This contradicts part (a), where we showed $\text{Hom}(M, M') = 0$. Therefore

$$M \not\cong M' \oplus M''.$$

□

(c) Show that there is a short exact sequence:

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0.$$

Proof. We construct explicit morphisms $\iota: M' \rightarrow M$ and $p: M \rightarrow M''$ such that

$$0 \longrightarrow M' \xrightarrow{\iota} M \xrightarrow{p} M'' \longrightarrow 0$$

is exact.

Definition of $\iota: M' \rightarrow M$. Since $M'(1) = M'(2) = M'(4) = 0$, we set

$$\iota_1 = 0: 0 \rightarrow k, \quad \iota_2 = 0: 0 \rightarrow k, \quad \iota_4 = 0: 0 \rightarrow k,$$

and define

$$\iota_3 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}: k \rightarrow k^2.$$

Since all maps of M' are zero, for each arrow $\delta: s \rightarrow 3$, the commutativity condition

$$\iota_3 \circ M'(\delta) = M(\delta) \circ \iota_s$$

becomes $0 = 0$, which is automatic. Moreover, ι_3 is injective (since $\begin{pmatrix} 1 \\ -1 \end{pmatrix} \neq 0$), so ι is injective.

Definition of $p: M \rightarrow M''$. Define

$$p_1 = 1: k \rightarrow k, \quad p_2 = 1: k \rightarrow k, \quad p_3 = \begin{pmatrix} 1 & 1 \end{pmatrix}: k^2 \rightarrow k, \quad p_4 = 2: k \rightarrow k.$$

We verify commutativity at each arrow.

At $\alpha: 1 \rightarrow 3$:

$$p_3 \circ M(\alpha) = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1 = 1 \cdot 1 = M''(\alpha) \circ p_1.$$

At $\beta: 2 \rightarrow 3$:

$$p_3 \circ M(\beta) = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 1 = 1 \cdot 1 = M''(\beta) \circ p_2.$$

At $\gamma: 4 \rightarrow 3$:

$$p_3 \circ M(\gamma) = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 2 = 1 \cdot 2 = M''(\gamma) \circ p_4.$$

Each p_i is a nonzero scalar, so p is surjective at every vertex.

Exactness at M .

First, $p \circ \iota = 0$: at vertex 3,

$$p_3 \circ \iota_3 = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 1 + (-1) = 0.$$

At other vertices, $\iota_i = 0$, so $(p \circ \iota)_i = p_i \circ 0 = 0$.

Next, we verify $\ker p = \text{im } \iota$ at each vertex:

- Vertices 1, 2, 4: each p_i is a nonzero scalar (hence injective), so $\ker p_i = 0$. Also $\iota_i = 0$, so $\text{im } \iota_i = 0$.
- Vertex 3:

$$\ker p_3 = \{(x, y) \in k^2 : x + y = 0\} = \left\langle \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\rangle = \text{im } \iota_3.$$

Therefore

$$0 \longrightarrow M' \xrightarrow{\iota} M \xrightarrow{p} M'' \longrightarrow 0$$

is a short exact sequence. □

6. Let Q be the quiver

$$1 \xrightleftharpoons[\beta]{\alpha} 2$$

For any $\lambda \in k \cup \{\infty\}$, define M_λ to be the representation:

$$M_\lambda: \quad k \xrightleftharpoons[1]{\lambda} k \quad \text{if } \lambda \in k; \quad M_\infty: \quad k \xrightleftharpoons[0]{1} k \quad \text{if } \lambda = \infty.$$

(a) Show that each M_λ is indecomposable.

Proof. Each M_λ has dimension vector

$$\underline{\dim} M_\lambda = (1, 1).$$

Suppose for contradiction that

$$M_\lambda = N_1 \oplus N_2$$

is a nontrivial decomposition (both $N_1, N_2 \neq 0$). Since $\underline{\dim} N_1 + \underline{\dim} N_2 = (1, 1)$, the only possibility is

$$\underline{\dim} N_1 = (1, 0), \quad \underline{\dim} N_2 = (0, 1)$$

(or vice versa). The summand N_2 with $N_2(1) = 0$ and $N_2(2) = k$ must be a subrepresentation. Since both arrows go from vertex 2 to vertex 1, this requires

$$M_\lambda(\alpha)(k) \subseteq N_2(1) = 0 \quad \text{and} \quad M_\lambda(\beta)(k) \subseteq N_2(1) = 0.$$

For $\lambda \in k$: the map $M_\lambda(\beta) = 1: k \rightarrow k$ is nonzero, so

$$M_\lambda(\beta)(k) = k \neq 0.$$

This is a contradiction.

For $\lambda = \infty$: the map $M_\infty(\alpha) = 1: k \rightarrow k$ is nonzero, so

$$M_\infty(\alpha)(k) = k \neq 0.$$

This is again a contradiction.

In both cases, no such subrepresentation N_2 exists, so M_λ is indecomposable. $\square$

- (b) Show that $M_\lambda \cong M_\mu$ if and only if $\lambda = \mu$. In particular, the number of indecomposable representations depends on the choice of the field k .

Proof. ($\Leftarrow$) If $\lambda = \mu$, the identity maps $f_1 = f_2 = \text{id}_k$ give an isomorphism.

($\Rightarrow$) Let $f = (f_1, f_2): M_\lambda \rightarrow M_\mu$ be an isomorphism, with

$$f_1 = a \in k^*, \quad f_2 = b \in k^*.$$

The commutativity conditions for the two arrows $\alpha, \beta: 2 \rightarrow 1$ are

$$f_1 \circ M_\lambda(\alpha) = M_\mu(\alpha) \circ f_2, \quad f_1 \circ M_\lambda(\beta) = M_\mu(\beta) \circ f_2.$$

Case 1: $\lambda, \mu \in k$. Then $M_\lambda(\alpha) = \lambda$, $M_\lambda(\beta) = 1$, $M_\mu(\alpha) = \mu$, $M_\mu(\beta) = 1$. The conditions become

$$a\lambda = \mu b, \quad a \cdot 1 = 1 \cdot b.$$

From the second equation, $a = b$. Substituting into the first:

$$a\lambda = \mu a.$$

Since $a \neq 0$, we can divide both sides by a to get $\lambda = \mu$.

Case 2: $\lambda \in k$, $\mu = \infty$. Then $M_\lambda(\beta) = 1$ and $M_\infty(\beta) = 0$. The commutativity at β gives

$$a \cdot 1 = 0 \cdot b = 0,$$

so $a = 0$, contradicting $a \in k^*$. Hence no isomorphism exists.

The case $\lambda = \infty$, $\mu \in k$ is symmetric.

Case 3: $\lambda = \mu = \infty$. Trivially $\lambda = \mu$.

In all cases, $M_\lambda \cong M_\mu$ implies $\lambda = \mu$. Since

$$|k \cup \{\infty\}| = |k| + 1,$$

the number of pairwise non-isomorphic indecomposable representations of this form depends on the cardinality of the field k . $\square$

(c) Show that $\text{Hom}(M_\lambda, M_\mu) = 0$ if $\lambda \neq \mu$.

Proof. Let $f = (f_1, f_2): M_\lambda \rightarrow M_\mu$ be a morphism, with $f_1 = a$ and $f_2 = b$ (scalars). The commutativity conditions are

$$f_1 \circ M_\lambda(\alpha) = M_\mu(\alpha) \circ f_2, \quad f_1 \circ M_\lambda(\beta) = M_\mu(\beta) \circ f_2.$$

Case 1: $\lambda, \mu \in k$. The conditions become

$$a\lambda = \mu b, \quad a = b.$$

Substituting $b = a$ into the first equation:

$$a\lambda = \mu a \implies a(\lambda - \mu) = 0.$$

Since $\lambda \neq \mu$, we have $\lambda - \mu \neq 0$, so $a = 0$. Then $b = a = 0$.

Case 2: $\lambda \in k, \mu = \infty$. Then $M_\lambda(\beta) = 1, M_\infty(\beta) = 0$. The condition at β gives

$$a \cdot 1 = 0 \cdot b = 0,$$

so $a = 0$. The condition at α then gives

$$0 \cdot \lambda = 1 \cdot b \implies b = 0.$$

Case 3: $\lambda = \infty, \mu \in k$. Then $M_\infty(\beta) = 0, M_\mu(\beta) = 1$. The condition at β gives

$$a \cdot 0 = 1 \cdot b \implies b = 0.$$

The condition at α gives

$$a \cdot 1 = \mu \cdot 0 = 0 \implies a = 0.$$

In every case, $f_1 = f_2 = 0$, so $f = 0$. Therefore $\text{Hom}(M_\lambda, M_\mu) = 0$. $\square$

(d) Show that for each λ there is a short exact sequence

$$0 \longrightarrow S(1) \longrightarrow M_\lambda \longrightarrow S(2) \longrightarrow 0,$$

where $S(1)$ and $S(2)$ are the representations

$$S(1): \quad k \xleftarrow{\quad} 0 \qquad S(2): \quad 0 \xleftarrow{\quad} k$$

Proof. Define $\iota: S(1) \rightarrow M_\lambda$ and $p: M_\lambda \rightarrow S(2)$ by

$$\begin{aligned}\iota_1 &= \text{id}_k: k \rightarrow k, & \iota_2 &= 0: 0 \rightarrow k, \\ p_1 &= 0: k \rightarrow 0, & p_2 &= \text{id}_k: k \rightarrow k.\end{aligned}$$

Morphism verification. Both arrows α, β go from vertex 2 to vertex 1. For ι , the commutativity diagram at each arrow $\delta \in \{\alpha, \beta\}$ is

$$\begin{array}{ccc} S(1)(2) = 0 & \xrightarrow{0} & S(1)(1) = k \\ \iota_2=0 \downarrow & & \downarrow \iota_1=\text{id}_k \\ M_\lambda(2) = k & \xrightarrow{M_\lambda(\delta)} & M_\lambda(1) = k \end{array}$$

Both compositions give 0, so the diagram commutes.

For p , the commutativity diagram at each arrow δ is

$$\begin{array}{ccc} M_\lambda(2) = k & \xrightarrow{M_\lambda(\delta)} & M_\lambda(1) = k \\ p_2=\text{id}_k \downarrow & & \downarrow p_1=0 \\ S(2)(2) = k & \xrightarrow{0} & S(2)(1) = 0 \end{array}$$

Both compositions give 0, so the diagram commutes.

Exactness.

ι is injective: $\iota_1 = \text{id}_k$ is injective.

p is surjective: $p_2 = \text{id}_k$ is surjective.

Exactness at M_λ : we verify $\ker p = \text{im } \iota$ at each vertex.

- Vertex 1:

$$\ker(p_1: k \rightarrow 0) = k = \text{im}(\iota_1) = \text{im}(\text{id}_k) = k.$$

- Vertex 2:

$$\ker(p_2) = \ker(\text{id}_k) = 0 = \text{im}(\iota_2) = \text{im}(0) = 0.$$

Hence

$$0 \longrightarrow S(1) \xrightarrow{\iota} M_\lambda \xrightarrow{p} S(2) \longrightarrow 0$$

is a short exact sequence. □

- 7.** Let $f: M \rightarrow N$ be a morphism of representations. Prove that the definition of $\text{coker } f$ is equivalent to the one defined in classroom through universal property (otherwise, see Remark 1.6 in Schiffler's book).

Proof. Let

$$f = (f_i): M = (M_i, \varphi_\alpha) \rightarrow N = (N_i, \psi_\alpha)$$

be a morphism of representations. For each vertex i , define

$$H_i := N_i / f_i(M_i).$$

For each arrow $\alpha : i \rightarrow j$, define

$$\chi_\alpha(n_i + f_i(M_i)) := \psi_\alpha(n_i) + f_j(M_j).$$

Hence

$$H = (H_i, \chi_\alpha)$$

is a representation.

Now let

$$g = (g_i) : N \rightarrow H$$

be the natural quotient morphism, where

$$g_i : N_i \rightarrow N_i/f_i(M_i), \quad g_i(n_i) = n_i + f_i(M_i).$$

Then for every vertex i and every $m_i \in M_i$,

$$(g_i \circ f_i)(m_i) = f_i(m_i) + f_i(M_i) = 0.$$

Thus

$$g \circ f = 0.$$

We now prove the universal property. Let

$$h = (h_i) : N \rightarrow Y = (Y_i, \eta_\alpha)$$

be a morphism such that

$$h \circ f = 0.$$

Then for each vertex i ,

$$h_i \circ f_i = 0.$$

Hence h_i vanishes on $f_i(M_i)$, so it factors uniquely through the quotient $N_i/f_i(M_i)$.

Therefore there exists a unique linear map

$$u_i : H_i = N_i/f_i(M_i) \rightarrow Y_i$$

such that

$$u_i(n_i + f_i(M_i)) := h_i(n_i).$$

This is well-defined because if

$$n_i + f_i(M_i) = n'_i + f_i(M_i),$$

then $n_i - n'_i = f_i(m_i)$ for some $m_i \in M_i$, and hence

$$h_i(n_i) - h_i(n'_i) = h_i(f_i(m_i)) = 0.$$

By construction,

$$u_i \circ g_i = h_i$$

for every vertex i .

It remains to check that $u = (u_i) : H \rightarrow Y$ is a morphism of representations. Let $\alpha : i \rightarrow j$ be an arrow and let $n_i + f_i(M_i) \in H_i$. Then

$$\begin{aligned} \eta_\alpha(u_i(n_i + f_i(M_i))) &= \eta_\alpha(h_i(n_i)) = h_j(\psi_\alpha(n_i)) = u_j(\psi_\alpha(n_i) + f_j(M_j)) \\ &= u_j(\chi_\alpha(n_i + f_i(M_i))). \end{aligned}$$

Therefore

$$\eta_\alpha \circ u_i = u_j \circ \chi_\alpha,$$

so u is a morphism of representations.

Finally, uniqueness is clear: if

$$u' = (u'_i) : H \rightarrow Y$$

also satisfies

$$u' \circ g = h,$$

then for each i ,

$$u'_i(n_i + f_i(M_i)) = u'_i(g_i(n_i)) = h_i(n_i) = u_i(n_i + f_i(M_i)).$$

Thus $u'_i = u_i$ for all i , so $u' = u$.

Therefore the quotient representation

$$(N_i/f_i(M_i), \chi_\alpha)$$

with the natural projection $g : N \rightarrow H$ satisfies the universal property of the cokernel.

On the other hand, suppose

$$g = (g_i) : N = (N_i, \psi_\alpha) \rightarrow H = (H_i, \chi_\alpha)$$

is a cokernel of

$$f = (f_i) : M = (M_i, \varphi_\alpha) \rightarrow N = (N_i, \psi_\alpha)$$

in the sense of the universal property.

Let

$$C := (N_i/f_i(M_i), \bar{\psi}_\alpha)$$

be the pointwise cokernel representation, where

$$\bar{\psi}_\alpha(n_i + f_i(M_i)) := \psi_\alpha(n_i) + f_j(M_j).$$

Let

$$\pi = (\pi_i) : N \rightarrow C$$

be the natural quotient morphism. As shown above,

$$\pi \circ f = 0.$$

Since $g : N \rightarrow H$ is a cokernel of f , applying its universal property to the morphism

$$\pi : N \rightarrow C$$

with $\pi \circ f = 0$, there exists a unique morphism

$$a : H \rightarrow C$$

such that

$$a \circ g = \pi.$$

On the other hand, since $g \circ f = 0$, for each vertex i we have

$$g_i \circ f_i = 0.$$

Thus g_i vanishes on $f_i(M_i)$, so it factors uniquely through the quotient

$$N_i/f_i(M_i).$$

Hence for each i there exists a unique linear map

$$b_i : C_i = N_i/f_i(M_i) \rightarrow H_i$$

given by

$$b_i(n_i + f_i(M_i)) := g_i(n_i).$$

These maps assemble into a morphism

$$b : C \rightarrow H,$$

because for any arrow $\alpha : i \rightarrow j$,

$$\begin{aligned} \chi_\alpha(b_i(n_i + f_i(M_i))) &= \chi_\alpha(g_i(n_i)) \\ &= g_j(\psi_\alpha(n_i)) \\ &= b_j(\psi_\alpha(n_i) + f_j(M_j)) \\ &= b_j(\overline{\psi}_\alpha(n_i + f_i(M_i))). \end{aligned}$$

So

$$b \circ \pi = g.$$

We now show that a and b are inverse isomorphisms.

First,

$$(a \circ b) \circ \pi = a \circ (b \circ \pi) = a \circ g = \pi.$$

Also,

$$\text{id}_C \circ \pi = \pi.$$

Since $\pi : N \rightarrow C$ is a cokernel of f , the uniqueness part of its universal property implies

$$a \circ b = \text{id}_C.$$

Similarly,

$$(b \circ a) \circ g = b \circ (a \circ g) = b \circ \pi = g,$$

and also

$$\text{id}_H \circ g = g.$$

Since $g : N \rightarrow H$ is a cokernel of f , the uniqueness part of its universal property implies

$$b \circ a = \text{id}_H.$$

Therefore a and b are inverse isomorphisms, and hence

$$H \cong C = (N_i/f_i(M_i), \overline{\psi}_\alpha).$$

In particular, for each vertex i we have

$$H_i \cong N_i/f_i(M_i),$$

and under these identifications, for every arrow $\alpha : i \rightarrow j$,

$$\chi_\alpha = a_j^{-1} \circ \overline{\psi}_\alpha \circ a_i.$$

Thus any cokernel given by the universal property is isomorphic to the pointwise quotient representation

$$(N_i/f_i(M_i), \psi_\alpha \text{ induced on the quotients}).$$

□

8. Let M, M' and N be representations of Q and let $f : M \rightarrow N, g : M' \rightarrow N$ be morphisms. Define the *fiber product* (or *pull back*) of f and g as

$$X = \{(a, b) \mid a \in M, b \in M', \text{ such that } f(a) = g(b)\},$$

and define the projections $\pi_1(a, b) = a$ and $\pi_2(a, b) = b$.

- (a) Show that X is a subrepresentation of $M \oplus M'$ and that there is a commutative

diagram:

$$\begin{array}{ccc} X & \xrightarrow{\pi_2} & M' \\ \pi_1 \downarrow & & \downarrow g \\ M & \xrightarrow{f} & N \end{array}$$

Proof. **X is a subrepresentation of $M \oplus M'$.**

At each vertex i , define

$$X_i = \{(a, b) \in M_i \oplus M'_i : f_i(a) = g_i(b)\}.$$

This is the kernel of the linear map

$$M_i \oplus M'_i \longrightarrow N_i, \quad (a, b) \longmapsto f_i(a) - g_i(b),$$

so X_i is a linear subspace of $M_i \oplus M'_i$.

We now show that for any arrow $\alpha: i \rightarrow j$, the map $(M \oplus M')(\alpha)$ sends X_i into X_j . Let $(a, b) \in X_i$, so

$$f_i(a) = g_i(b).$$

We need to show $(M(\alpha)(a), M'(\alpha)(b)) \in X_j$, i.e., $f_j(M(\alpha)(a)) = g_j(M'(\alpha)(b))$.

Using the commutativity of f and g :

$$f_j(M(\alpha)(a)) = N(\alpha)(f_i(a)) = N(\alpha)(g_i(b)) = g_j(M'(\alpha)(b)).$$

Hence X is stable under the maps of $M \oplus M'$ and is therefore a subrepresentation.

Commutativity of the diagram.

For any $(a, b) \in X_i$, we have

$$(f \circ \pi_1)_i(a, b) = f_i(\pi_1(a, b)) = f_i(a) = g_i(b) = g_i(\pi_2(a, b)) = (g \circ \pi_2)_i(a, b).$$

Hence $f \circ \pi_1 = g \circ \pi_2$, and the diagram

$$\begin{array}{ccc} X & \xrightarrow{\pi_2} & M' \\ \pi_1 \downarrow & & \downarrow g \\ M & \xrightarrow{f} & N \end{array}$$

commutes. □

(b) Show that if f is injective then π_2 is injective.

Proof. Suppose f is injective, i.e., f_i is injective for each vertex i .

Let $(a, b) \in X_i$ with

$$\pi_2(a, b) = b = 0.$$

Since $(a, b) \in X_i$, we have $f_i(a) = g_i(b)$. Substituting $b = 0$:

$$f_i(a) = g_i(0) = 0.$$

Since f_i is injective, it follows that

$$a = 0.$$

Hence $(a, b) = (0, 0)$, so $\ker(\pi_2)_i = 0$ at every vertex i . Therefore π_2 is injective. $\square$

(c) Show that if f is surjective then π_2 is surjective.

Proof. Suppose f is surjective, i.e., f_i is surjective for each vertex i .

Let $b \in M'_i$. We need to find $a \in M_i$ such that $(a, b) \in X_i$, i.e., such that

$$f_i(a) = g_i(b).$$

The element $g_i(b) \in N_i$. Since f_i is surjective, we have

$$N_i = \text{im}(f_i),$$

so there exists $a \in M_i$ with $f_i(a) = g_i(b)$. Then $(a, b) \in X_i$ by definition, and

$$\pi_2(a, b) = b.$$

Since $b \in M'_i$ was arbitrary, π_2 is surjective at each vertex. $\square$

(d) Suppose $0 \rightarrow L \xrightarrow{h} M \xrightarrow{f} N \rightarrow 0$ is a short exact sequence, define $h': L \rightarrow X$ by $h'(n) = (h(n), 0)$. Show that the following diagram is commutative with exact rows:

$$\begin{array}{ccccccc} 0 & \longrightarrow & L & \xrightarrow{h'} & X & \xrightarrow{\pi_2} & M' \longrightarrow 0 \\ & & \downarrow 1_L & & \downarrow \pi_1 & & \downarrow g \\ 0 & \longrightarrow & L & \xrightarrow{h} & M & \xrightarrow{f} & N \longrightarrow 0 \end{array}$$

Proof. **h' is well-defined.**

For $n \in L_i$, we must check that $(h_i(n), 0) \in X_i$, i.e., that

$$f_i(h_i(n)) = g_i(0).$$

By exactness of the bottom row at M , we have $f \circ h = 0$, so

$$f_i(h_i(n)) = 0 = g_i(0).$$

Hence $h'_i(n) = (h_i(n), 0) \in X_i$.

h' is a morphism of representations.

For each arrow $\alpha: i \rightarrow j$ and $n \in L_i$:

$$\begin{aligned} X(\alpha)(h'_i(n)) &= X(\alpha)(h_i(n), 0) \\ &= (M(\alpha)(h_i(n)), M'(\alpha)(0)) \\ &= (h_j(L(\alpha)(n)), 0) \\ &= h'_j(L(\alpha)(n)), \end{aligned}$$

where the third equality uses the commutativity of h , i.e., $M(\alpha) \circ h_i = h_j \circ L(\alpha)$.

Commutativity of the diagram.

Left square: for each $n \in L_i$,

$$(\pi_1 \circ h')_i(n) = \pi_1(h_i(n), 0) = h_i(n) = (h \circ 1_L)_i(n).$$

So $\pi_1 \circ h' = h \circ 1_L$.

Right square: $f \circ \pi_1 = g \circ \pi_2$ was established in part (a).

Exactness of the top row.

h' is injective: suppose

$$h'_i(n) = (h_i(n), 0) = (0, 0).$$

Then $h_i(n) = 0$. Since h is injective (from the exactness of the bottom row),

$$n = 0.$$

Hence h' is injective.

π_2 is surjective: since f is surjective (from the bottom row), this follows from part (c).

$\text{im}(h') = \ker(\pi_2)$:

($\supseteq$) For any $n \in L_i$,

$$\pi_2(h'_i(n)) = \pi_2(h_i(n), 0) = 0,$$

so $\text{im}(h'_i) \subseteq \ker(\pi_2)_i$.

($\subseteq$) Let $(a, b) \in X_i$ with $\pi_2(a, b) = b = 0$. Then

$$f_i(a) = g_i(0) = 0,$$

so $a \in \ker f_i$. By exactness of the bottom row at M ,

$$\ker f_i = \text{im } h_i.$$

Hence there exists $n \in L_i$ with $a = h_i(n)$, and then

$$(a, 0) = (h_i(n), 0) = h'_i(n) \in \text{im}(h'_i).$$

Therefore $\text{im}(h') = \ker(\pi_2)$, and the top row is exact. □

9. Let $f: L \rightarrow M$ be a morphism. Then a *cokernel* of f is a morphism $g: M \rightarrow N$ such that $gf = 0$, and given any morphism $v: M \rightarrow X$ such that $vf = 0$, there is a unique morphism $u: N \rightarrow X$ such that $ug = v$:

$$\begin{array}{ccccc} L & \xrightarrow{f} & M & \xrightarrow{g} & N \\ & & \downarrow v & \swarrow \exists u & \\ & & X & & \end{array}$$

Prove that the two definitions of cokernel agree.

Proof. Let

$$f = (f_i): L = (L_i, \varphi_\alpha) \rightarrow M = (M_i, \psi_\alpha)$$

be a morphism of representations.

We first show that the pointwise cokernel satisfies the universal property. For each vertex $i \in Q_0$, set

$$H_i := M_i / f_i(L_i),$$

and let

$$\pi_i: M_i \rightarrow H_i$$

be the canonical quotient map. For an arrow $\alpha: i \rightarrow j$, define

$$\bar{\psi}_\alpha: H_i \rightarrow H_j$$

by

$$\bar{\psi}_\alpha(m_i + f_i(L_i)) := \psi_\alpha(m_i) + f_j(L_j).$$

This is well defined because f is a morphism, so

$$\psi_\alpha(f_i(L_i)) = f_j(\varphi_\alpha(L_i)) \subseteq f_j(L_j).$$

Hence

$$H = (H_i, \bar{\psi}_\alpha)$$

is a representation, and $\pi = (\pi_i): M \rightarrow H$ is a morphism.

Moreover,

$$(\pi \circ f)_i = \pi_i \circ f_i = 0$$

for every vertex i , since $f_i(L_i) \subseteq \ker \pi_i$. Thus

$$\pi \circ f = 0.$$

Now let $v = (v_i): M \rightarrow X = (X_i, \eta_\alpha)$ be a morphism such that

$$v \circ f = 0.$$

Then for every vertex i we have

$$v_i \circ f_i = 0,$$

so v_i vanishes on $f_i(L_i)$. By the universal property of the quotient vector space $H_i = M_i/f_i(L_i)$, there exists a unique linear map

$$u_i: H_i \rightarrow X_i$$

such that

$$u_i \circ \pi_i = v_i.$$

We claim that the family (u_i) defines a morphism of representations

$$u: H \rightarrow X.$$

Indeed, let $\alpha: i \rightarrow j$ be an arrow. Then

$$\eta_\alpha \circ u_i \circ \pi_i = \eta_\alpha \circ v_i = v_j \circ \psi_\alpha = u_j \circ \pi_j \circ \psi_\alpha = u_j \circ \bar{\psi}_\alpha \circ \pi_i.$$

Since π_i is surjective, it follows that

$$\eta_\alpha \circ u_i = u_j \circ \bar{\psi}_\alpha.$$

Therefore $u: H \rightarrow X$ is a morphism, and by construction

$$u \circ \pi = v.$$

To prove uniqueness, suppose $u': H \rightarrow X$ is another morphism such that

$$u' \circ \pi = v.$$

Then for every vertex i ,

$$u'_i \circ \pi_i = v_i = u_i \circ \pi_i.$$

Since π_i is surjective, we get $u'_i = u_i$. Hence $u' = u$. Thus $\pi: M \rightarrow H$ satisfies the universal property of the cokernel of f .

Next, suppose that

$$g: M \rightarrow N$$

is any cokernel of f in the sense of the universal property. Since $\pi \circ f = 0$, the universal property of g yields a unique morphism

$$a: N \rightarrow H$$

such that

$$a \circ g = \pi.$$

Similarly, since $g \circ f = 0$, the universal property of π yields a unique morphism

$$b: H \rightarrow N$$

such that

$$b \circ \pi = g.$$

We now show that a and b are inverse isomorphisms. First,

$$(a \circ b) \circ \pi = a \circ (b \circ \pi) = a \circ g = \pi = \text{id}_H \circ \pi.$$

Both $a \circ b$ and id_H are morphisms $H \rightarrow H$ whose composition with π is equal to π . By the universal property of π , such a morphism is unique. Therefore

$$a \circ b = \text{id}_H.$$

Similarly,

$$(b \circ a) \circ g = b \circ (a \circ g) = b \circ \pi = g = \text{id}_N \circ g.$$

Both $b \circ a$ and id_N are morphisms $N \rightarrow N$ whose composition with g is equal to g . By the universal property of g , such a morphism is unique. Hence

$$b \circ a = \text{id}_N.$$

Therefore a and b are inverse isomorphisms, so $N \cong H$. This proves that any cokernel in the universal-property sense is isomorphic to the pointwise cokernel. Hence the two definitions agree. $\square$

10. Let Q be a quiver and let

$$L \xrightarrow{f} M \xrightarrow{g} N \longrightarrow 0$$

be a sequence in $\text{Rep } Q$. Show that this sequence is exact if and only if for every representation $X \in \text{Rep } Q$, the sequence

$$0 \longrightarrow \text{Hom}(N, X) \xrightarrow{g^*} \text{Hom}(M, X) \xrightarrow{f^*} \text{Hom}(L, X)$$

is exact, where $g^*(h) = h \circ g$ and $f^*(k) = k \circ f$.

Proof. ($\Rightarrow$) Assume that

$$L \xrightarrow{f} M \xrightarrow{g} N \longrightarrow 0$$

is exact. Let $X \in \text{Rep } Q$ be arbitrary. We show that

$$0 \longrightarrow \text{Hom}(N, X) \xrightarrow{g^*} \text{Hom}(M, X) \xrightarrow{f^*} \text{Hom}(L, X)$$

is exact.

First, g^* is injective. Indeed, if $h, h' \in \text{Hom}(N, X)$ satisfy

$$g^*(h) = g^*(h'),$$

then

$$h \circ g = h' \circ g.$$

Since g is surjective, it follows that $h = h'$.

Next, for any $h \in \text{Hom}(N, X)$,

$$f^*(g^*(h)) = (h \circ g) \circ f = h \circ (g \circ f) = 0,$$

because $g \circ f = 0$. Hence

$$\text{im } g^* \subseteq \ker f^*.$$

Conversely, let $k \in \text{Hom}(M, X)$ satisfy

$$f^*(k) = k \circ f = 0.$$

Since the sequence

$$L \xrightarrow{f} M \xrightarrow{g} N \longrightarrow 0$$

is exact, the morphism g is a cokernel of f . Therefore, by the universal property of the cokernel, there exists a unique morphism

$$u : N \rightarrow X$$

such that

$$u \circ g = k.$$

Thus $k = g^*(u)$, and so

$$\ker f^* \subseteq \text{im } g^*.$$

Therefore

$$\ker f^* = \text{im } g^*,$$

and the induced sequence is exact.

($\Leftarrow$) Assume that for every $X \in \text{Rep } Q$, the sequence

$$0 \longrightarrow \text{Hom}(N, X) \xrightarrow{g^*} \text{Hom}(M, X) \xrightarrow{f^*} \text{Hom}(L, X)$$

is exact. We show that

$$L \xrightarrow{f} M \xrightarrow{g} N \longrightarrow 0$$

is exact.

First, we prove that $g \circ f = 0$. Take $X = N$. Since

$$f^* \circ g^* = 0,$$

evaluating at $\text{id}_N \in \text{Hom}(N, N)$ gives

$$0 = f^*(g^*(\text{id}_N)) = \text{id}_N \circ g \circ f = g \circ f.$$

Next, we show that g is surjective. Let

$$\pi : N \rightarrow \text{coker } g$$

be the cokernel morphism of g . Since $\pi \circ g = 0$, we have

$$g^*(\pi) = \pi \circ g = 0.$$

By exactness, g^* is injective, hence $\pi = 0$. Therefore

$$\text{coker } g = 0,$$

so g is surjective.

Finally, we show that $\ker g = \text{im } f$. Since $g \circ f = 0$, we have

$$\text{im } f \subseteq \ker g.$$

For the reverse inclusion, let

$$\gamma : M \rightarrow \text{coker } f$$

be the cokernel morphism of f . Since $\gamma \circ f = 0$, we have

$$f^*(\gamma) = 0.$$

By exactness of

$$0 \longrightarrow \text{Hom}(N, \text{coker } f) \xrightarrow{g^*} \text{Hom}(M, \text{coker } f) \xrightarrow{f^*} \text{Hom}(L, \text{coker } f),$$

it follows that $\gamma \in \text{im } g^*$. Hence there exists a morphism

$$u : N \rightarrow \text{coker } f$$

such that

$$\gamma = u \circ g.$$

Therefore

$$\ker g \subseteq \ker \gamma.$$

Since γ is the cokernel morphism of f , we have

$$\ker \gamma = \text{im } f.$$

Thus

$$\ker g \subseteq \operatorname{im} f.$$

Combining this with the reverse inclusion, we get

$$\ker g = \operatorname{im} f.$$

Hence

$$L \xrightarrow{f} M \xrightarrow{g} N \longrightarrow 0$$

is exact. □

- 11.** Let $g: M \rightarrow N$ be a surjective morphism between representations of Q , and let $P(i)$ be the projective representation at vertex i . Show that the map

$$g_*: \operatorname{Hom}(P(i), M) \rightarrow \operatorname{Hom}(P(i), N), \quad \varphi \mapsto g \circ \varphi$$

is surjective.

Proof. Let

$$g_*: \operatorname{Hom}(P(i), M) \longrightarrow \operatorname{Hom}(P(i), N), \quad \varphi \mapsto g \circ \varphi.$$

We must show that g_* is surjective.

Recall that for every representation $X \in \operatorname{Rep} Q$, there is a natural isomorphism

$$\Phi_X: \operatorname{Hom}(P(i), X) \xrightarrow{\sim} X_i, \quad f \mapsto f(e_i),$$

where e_i denotes the trivial path at the vertex i .

We claim that the following diagram commutes:

$$\begin{array}{ccc} \operatorname{Hom}(P(i), M) & \xrightarrow[\sim]{\Phi_M} & M_i \\ g_* \downarrow & & \downarrow g_i \\ \operatorname{Hom}(P(i), N) & \xrightarrow[\sim]{\Phi_N} & N_i \end{array}$$

Indeed, for any $f \in \operatorname{Hom}(P(i), M)$, we have

$$\Phi_N(g_*(f)) = \Phi_N(g \circ f) = (g \circ f)_i(e_i) = g_i(f_i(e_i)) = g_i(\Phi_M(f)).$$

Thus

$$\Phi_N \circ g_* = g_i \circ \Phi_M.$$

Since $g: M \rightarrow N$ is surjective, each component map

$$g_j: M_j \rightarrow N_j$$

is surjective. In particular, $g_i: M_i \rightarrow N_i$ is surjective. Because Φ_M and Φ_N are isomorphisms, the commutative diagram above shows that g_* corresponds to the surjective

map g_i . Hence g_* is surjective.

Therefore every morphism $P(i) \rightarrow N$ lifts to a morphism $P(i) \rightarrow M$, as required. $\square$

- 12.** Let $g: L \rightarrow M$ be an injective morphism between representations of Q , and let $I(i)$ be the injective representation at vertex i . Show that the map

$$g^*: \text{Hom}(M, I(i)) \rightarrow \text{Hom}(L, I(i)), \quad \psi \mapsto \psi \circ g$$

is surjective.

Proof. For every representation X , there is a natural isomorphism

$$\Psi_X: \text{Hom}(X, I(i)) \xrightarrow{\sim} X_i^*,$$

where $X_i^* = \text{Hom}_k(X_i, k)$. Concretely, after identifying

$$I(i)_i \cong k$$

via the trivial path e_i , a morphism $f: X \rightarrow I(i)$ is sent to the linear functional

$$\Psi_X(f) = f_i \in \text{Hom}_k(X_i, k) = X_i^*.$$

Under these identifications, the map

$$g^*: \text{Hom}(M, I(i)) \rightarrow \text{Hom}(L, I(i)), \quad \psi \mapsto \psi \circ g,$$

corresponds to the dual map

$$g_i^*: M_i^* \rightarrow L_i^*, \quad \lambda \mapsto \lambda \circ g_i.$$

Indeed, for every $\psi \in \text{Hom}(M, I(i))$, we have

$$\Psi_L(g^*(\psi)) = \Psi_L(\psi \circ g) = (\psi \circ g)_i = \psi_i \circ g_i = g_i^*(\psi_i) = g_i^*(\Psi_M(\psi)).$$

Hence the following diagram commutes:

$$\begin{array}{ccc} \text{Hom}(M, I(i)) & \xrightarrow[\sim]{\Psi_M} & M_i^* \\ g^* \downarrow & & \downarrow g_i^* \\ \text{Hom}(L, I(i)) & \xrightarrow[\sim]{\Psi_L} & L_i^* \end{array}$$

Now $g: L \rightarrow M$ is injective, so each component

$$g_j: L_j \rightarrow M_j$$

is injective; in particular, g_i is injective. Since we are working with finite-dimensional

representations, L_i and M_i are finite-dimensional vector spaces. Therefore the dual map

$$g_i^*: M_i^* \rightarrow L_i^*$$

is surjective.

For completeness, let $\lambda \in L_i^*$. Since g_i is injective, it induces an isomorphism

$$L_i \xrightarrow{\sim} g_i(L_i) \subseteq M_i.$$

Thus λ defines a linear form on the subspace $g_i(L_i)$ by

$$g_i(x) \mapsto \lambda(x).$$

By linear algebra, this linear form extends to some $\mu \in M_i^*$. Then

$$g_i^*(\mu) = \mu \circ g_i = \lambda.$$

Hence g_i^* is surjective.

Since Ψ_M and Ψ_L are isomorphisms and the diagram above commutes, it follows that

$$g^*: \text{Hom}(M, I(i)) \rightarrow \text{Hom}(L, I(i))$$

is surjective. □

13. Let Q be the quiver

$$\begin{array}{ccccc} 1 & \xrightarrow{\quad} & 3 & \xrightarrow{\quad} & 4 \\ & \searrow & \nearrow & & \\ & 2 & & & \end{array}$$

Prove that

$$\begin{array}{ccccc} & & \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} & & \begin{bmatrix} a & b \\ c & d \end{bmatrix} \\ & & \downarrow & & \downarrow \\ P(1) \simeq k & \xrightarrow{\quad} & k^2 & \xrightarrow{\quad} & k^2 \\ & \searrow z & \nearrow \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} & & \end{array}$$

if and only if all four maps have maximal rank, that is,

$$ad - bc \neq 0, \quad (x_1, x_2) \neq (0, 0), \quad (y_1, y_2) \neq (0, 0), \quad z \neq 0$$

and the vectors

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

are linearly independent.

Proof. Let the arrows of Q be

$$\alpha: 1 \rightarrow 3, \quad \beta: 1 \rightarrow 2, \quad \gamma: 2 \rightarrow 3, \quad \delta: 3 \rightarrow 4,$$

and let M denote the representation

$$\begin{array}{ccccc} k & \xrightarrow{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}} & k^2 & \xrightarrow{\begin{bmatrix} a & b \\ c & d \end{bmatrix}} & k^2 \\ & \searrow z & \nearrow \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} & & \end{array}$$

that is,

$$M_\alpha = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad M_\beta = z, \quad M_\gamma = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}, \quad M_\delta = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Step 1: Description of $P(1)$. Since Q has no oriented cycles, the projective representation $P(1)$ is given at each vertex j by the k -vector space with basis the set of paths from 1 to j . Thus

$$P(1)_1 \cong k, \quad P(1)_2 \cong k, \quad P(1)_3 \cong k^2, \quad P(1)_4 \cong k^2,$$

because:

- there is one path $1 \rightarrow 1$, namely the trivial path e_1 ;
- there is one path $1 \rightarrow 2$, namely β ;
- there are two paths $1 \rightarrow 3$, namely α and $\gamma\beta$;
- there are two paths $1 \rightarrow 4$, namely $\delta\alpha$ and $\delta\gamma\beta$.

Choose the ordered bases

$$P(1)_1 = (e_1), \quad P(1)_2 = (\beta), \quad P(1)_3 = (\alpha, \gamma\beta), \quad P(1)_4 = (\delta\alpha, \delta\gamma\beta).$$

With respect to these bases, the structure maps of $P(1)$ are

$$P(1)_\beta = 1, \quad P(1)_\alpha = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad P(1)_\gamma = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad P(1)_\delta = I_2.$$

Hence $P(1)$ has the form

$$\begin{array}{ccccc}
 & & \begin{bmatrix} 1 \\ 0 \end{bmatrix} & & \\
 & & \downarrow & & \\
 P(1) \simeq & k & \xrightarrow{\quad} & k^2 & \xrightarrow{I_2} k^2 \\
 & \searrow 1 & & \nearrow \begin{bmatrix} 0 \\ 1 \end{bmatrix} & \\
 & & k & &
 \end{array}$$

In particular, all four maps have maximal rank, and the two vectors

$$P(1)_\delta P(1)_\alpha = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad P(1)_\delta P(1)_\gamma = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

are linearly independent.

Step 2: Necessity. Assume that $M \simeq P(1)$. Let

$$\phi = (\phi_1, \phi_2, \phi_3, \phi_4): P(1) \xrightarrow{\sim} M$$

be an isomorphism of representations.

Since isomorphisms preserve rank, and since the maps of $P(1)$ all have maximal rank, it follows immediately that the four maps of M also have maximal rank. Thus

$$ad - bc \neq 0, \quad (x_1, x_2) \neq (0, 0), \quad (y_1, y_2) \neq (0, 0), \quad z \neq 0.$$

It remains to show that

$$M_\delta M_\alpha \quad \text{and} \quad M_\delta M_\gamma$$

are linearly independent in k^2 .

Since $\phi_1, \phi_2: k \rightarrow k$ are automorphisms, there exist nonzero scalars

$$\phi_1 = \lambda_1, \quad \phi_2 = \lambda_2$$

with $\lambda_1, \lambda_2 \in k^\times$. The commutativity of the squares gives

$$\phi_3 P(1)_\alpha = M_\alpha \phi_1, \quad \phi_3 P(1)_\gamma = M_\gamma \phi_2, \quad \phi_4 P(1)_\delta = M_\delta \phi_3.$$

Therefore

$$M_\delta M_\alpha \lambda_1 = M_\delta \phi_3 P(1)_\alpha = \phi_4 P(1)_\delta P(1)_\alpha,$$

and similarly

$$M_\delta M_\gamma \lambda_2 = \phi_4 P(1)_\delta P(1)_\gamma.$$

Since ϕ_4 is invertible and $P(1)_\delta P(1)_\alpha, P(1)_\delta P(1)_\gamma$ are linearly independent, the vectors

$$\phi_4 P(1)_\delta P(1)_\alpha \quad \text{and} \quad \phi_4 P(1)_\delta P(1)_\gamma$$

are linearly independent. Multiplication by the nonzero scalars λ_1^{-1} and λ_2^{-1} does not affect linear independence, so

$$M_\delta M_\alpha \quad \text{and} \quad M_\delta M_\gamma$$

are linearly independent as required.

Step 3: Sufficiency. Conversely, assume that all four maps of M have maximal rank, namely

$$ad - bc \neq 0, \quad (x_1, x_2) \neq (0, 0), \quad (y_1, y_2) \neq (0, 0), \quad z \neq 0,$$

and assume moreover that the two vectors

$$M_\delta M_\alpha = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad M_\delta M_\gamma = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

are linearly independent.

Since $ad - bc \neq 0$, the matrix M_δ is invertible. Hence the linear independence of

$$M_\delta \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \text{and} \quad M_\delta \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

implies that

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

are linearly independent. In particular,

$$x_1 y_2 - x_2 y_1 \neq 0.$$

We now define maps

$$\begin{aligned} \phi_1 &= 1: k \rightarrow k, & \phi_2 &= z: k \rightarrow k, \\ \phi_3 &= \begin{bmatrix} x_1 & zy_1 \\ x_2 & zy_2 \end{bmatrix}: k^2 \rightarrow k^2, & \phi_4 &= M_\delta \phi_3: k^2 \rightarrow k^2. \end{aligned}$$

We claim that

$$\phi = (\phi_1, \phi_2, \phi_3, \phi_4): P(1) \rightarrow M$$

is an isomorphism of representations.

Indeed, we check the commutativity of the four arrow maps:

$$\phi_2 P(1)_\beta = z \cdot 1 = z = M_\beta \cdot 1 = M_\beta \phi_1,$$

$$\phi_3 P(1)_\alpha = \phi_3 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = M_\alpha \phi_1,$$

$$\phi_3 P(1)_\gamma = \phi_3 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = z \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = M_\gamma \phi_2,$$

and

$$\phi_4 P(1)_\delta = \phi_4 I_2 = \phi_4 = M_\delta \phi_3.$$

Thus ϕ is a morphism of representations.

Finally, each ϕ_i is invertible:

$$\phi_1 = 1 \neq 0, \quad \phi_2 = z \neq 0,$$

$$\det(\phi_3) = z(x_1 y_2 - x_2 y_1) \neq 0,$$

and

$$\det(\phi_4) = \det(M_\delta) \det(\phi_3) \neq 0.$$

Hence ϕ is an isomorphism, and therefore

$$M \simeq P(1).$$

This proves that

$$P(1) \simeq \begin{array}{ccccc} & & \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} & & \begin{bmatrix} a & b \\ c & d \end{bmatrix} \\ & & \xrightarrow{\quad} & & \xrightarrow{\quad} \\ k & \xrightarrow{\quad} & k^2 & \xrightarrow{\quad} & k^2 \\ & \searrow z & \nearrow \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} & & \end{array}$$

if and only if all four maps have maximal rank and the two vectors

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

are linearly independent. □

- 14.** Compute the indecomposable projective (resp. injective) representations $P(i)$ (resp. $I(i)$) for the following quivers:

$$Q_1: \quad 1 \xrightleftharpoons{\quad} 2 \xrightleftharpoons{\quad} 3 \qquad Q_2: \quad \begin{array}{ccccc} & & & \curvearrowright & \\ & & & \xrightarrow{\quad} & \\ 1 & \xrightarrow{\quad} & 2 & \xrightarrow{\quad} & 3 \\ & & \downarrow & \nwarrow & \\ & & 4 & & \end{array}$$

Proof. We describe the indecomposable projective and injective representations explicitly, up to isomorphism, by choosing natural path bases.

Recall that for a quiver Q and a vertex $i \in Q_0$,

$$P(i)_j \cong k\{\text{paths } i \rightarrow j\}, \quad I(i)_j \cong k\{\text{paths } j \rightarrow i\},$$

where $k\{S\}$ denotes the k -vector space with basis S . The linear map attached to an arrow is induced by composition of paths.

We use the convention that a product of arrows is read from right to left: for instance, $\alpha\gamma$ means “first γ , then α ”.

(1) The quiver Q_1 .

$$Q_1: \quad 1 \xleftarrow{\quad} 2 \xleftarrow{\quad} 3$$

Label the arrows by

$$\alpha, \beta: 2 \rightarrow 1, \quad \gamma, \delta: 3 \rightarrow 2.$$

Indecomposable projectives.

The projective $P(1)$. Since 1 is a sink, the only path starting at 1 is the trivial path e_1 . Hence

$$P(1) = S(1): \quad k \xleftarrow{\quad} 0 \xleftarrow{\quad} 0.$$

The projective $P(2)$. The paths starting at 2 are

$$e_2 \text{ (ending at 2),} \quad \alpha, \beta \text{ (ending at 1).}$$

Thus

$$P(2)_1 \cong k\{\alpha, \beta\} \cong k^2, \quad P(2)_2 \cong k\{e_2\} \cong k, \quad P(2)_3 = 0.$$

With respect to the ordered basis $\{\alpha, \beta\}$ at vertex 1, we obtain

$$P(2): \quad k^2 \begin{smallmatrix} (0) \\ \xleftarrow{\quad} \\ (1) \end{smallmatrix} k \xleftarrow{\quad} 0.$$

The projective $P(3)$. The paths starting at 3 are

$$e_3 \text{ (ending at 3),} \quad \gamma, \delta \text{ (ending at 2),}$$

and

$$\alpha\gamma, \alpha\delta, \beta\gamma, \beta\delta \text{ (ending at 1).}$$

Hence

$$P(3)_1 \cong k^4, \quad P(3)_2 \cong k^2, \quad P(3)_3 \cong k,$$

where we choose the ordered bases

$$P(3)_1: \{\alpha\gamma, \alpha\delta, \beta\gamma, \beta\delta\}, \quad P(3)_2: \{\gamma, \delta\}.$$

Then

$$P(3): \quad k^4 \begin{array}{c} \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \\ \hline \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} \end{array} k^2 \begin{array}{c} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ \hline \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{array} k .$$

Indecomposable injectives.

The injective $I(3)$. Since 3 is a source, the only path ending at 3 is the trivial path e_3 . Therefore

$$I(3) = S(3): \quad 0 \begin{array}{c} \longleftarrow \\ \hline \longleftarrow \end{array} 0 \begin{array}{c} \longleftarrow \\ \hline \longleftarrow \end{array} k .$$

The injective $I(2)$. The paths ending at 2 are

$$e_2 \text{ (starting at 2),} \quad \gamma, \delta \text{ (starting at 3).}$$

Thus

$$I(2)_1 = 0, \quad I(2)_2 \cong k, \quad I(2)_3 \cong k^2.$$

With respect to the ordered basis $\{\gamma, \delta\}$ at vertex 3, we get

$$I(2): \quad 0 \begin{array}{c} \longleftarrow \\ \hline \longleftarrow \end{array} k \begin{array}{c} \begin{pmatrix} 0,1 \\ \hline 1,0 \end{pmatrix} \\ \hline \begin{pmatrix} 0,1 \\ \hline 1,0 \end{pmatrix} \end{array} k^2 .$$

The injective $I(1)$. The paths ending at 1 are

$$e_1 \text{ (starting at 1),} \quad \alpha, \beta \text{ (starting at 2),}$$

and

$$\alpha\gamma, \alpha\delta, \beta\gamma, \beta\delta \text{ (starting at 3).}$$

Hence

$$I(1)_1 \cong k, \quad I(1)_2 \cong k^2, \quad I(1)_3 \cong k^4.$$

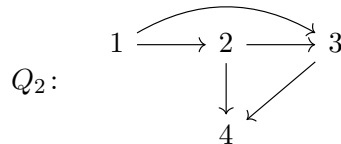
Choose the ordered bases

$$I(1)_2: \{\alpha, \beta\}, \quad I(1)_3: \{\alpha\gamma, \alpha\delta, \beta\gamma, \beta\delta\}.$$

Then

$$I(1): \quad k \begin{array}{c} \begin{pmatrix} 0,1 \\ \hline 1,0 \end{pmatrix} \\ \hline \begin{pmatrix} 0,1 \\ \hline 1,0 \end{pmatrix} \end{array} k^2 \begin{array}{c} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \\ \hline \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \end{array} k^4 .$$

(2) The quiver Q_2 .



Label the arrows by

$$\alpha: 1 \rightarrow 2, \quad \beta: 1 \rightarrow 3, \quad \gamma: 2 \rightarrow 3, \quad \delta: 2 \rightarrow 4, \quad \epsilon: 3 \rightarrow 4.$$

Indecomposable projectives.

The projective $P(4)$. Since 4 is a sink, one has

$$P(4) = S(4): \quad \begin{array}{ccccc} & & \curvearrowright & & \\ 0 & \longrightarrow & 0 & \longrightarrow & 0 \\ & & \downarrow & \nearrow & \\ & & k & & \end{array}.$$

The projective $P(3)$. The paths starting at 3 are

$$e_3 \text{ (ending at 3),} \quad \epsilon \text{ (ending at 4).}$$

Thus

$$P(3)_1 = 0, \quad P(3)_2 = 0, \quad P(3)_3 \cong k, \quad P(3)_4 \cong k,$$

and the arrow $\epsilon: 3 \rightarrow 4$ induces the identity map. Hence

$$P(3): \quad \begin{array}{ccccc} & & \curvearrowright & & \\ 0 & \longrightarrow & 0 & \longrightarrow & k \\ & & \downarrow & \nearrow & \\ & & k & & \end{array} \quad \begin{array}{c} \\ \\ 1 \end{array}.$$

The projective $P(2)$. The paths starting at 2 are

$$e_2 \text{ (ending at 2),} \quad \gamma \text{ (ending at 3),} \quad \delta, \epsilon\gamma \text{ (ending at 4).}$$

Therefore

$$P(2)_1 = 0, \quad P(2)_2 \cong k, \quad P(2)_3 \cong k, \quad P(2)_4 \cong k^2.$$

Using the ordered basis $\{\delta, \epsilon\gamma\}$ at vertex 4, we get

$$P(2): \quad \begin{array}{ccccc} & & \curvearrowright & & \\ 0 & \longrightarrow & k & \xrightarrow{1} & k \\ & & \downarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix} & \nearrow \begin{pmatrix} 0 \\ 1 \end{pmatrix} & \\ & & k^2 & & \end{array}.$$

The projective $P(1)$. The paths starting at 1 are

$$e_1 \text{ (ending at 1),} \quad \alpha \text{ (ending at 2),}$$

$$\beta, \gamma\alpha \text{ (ending at 3),} \quad \delta\alpha, \epsilon\beta, \epsilon\gamma\alpha \text{ (ending at 4).}$$

Hence

$$P(1)_1 \cong k, \quad P(1)_2 \cong k, \quad P(1)_3 \cong k^2, \quad P(1)_4 \cong k^3.$$

Choose the ordered bases

$$P(1)_3 : \{\beta, \gamma\alpha\}, \quad P(1)_4 : \{\delta\alpha, \epsilon\beta, \epsilon\gamma\alpha\}.$$

Then

$$P(1): \begin{array}{ccccc} & & \begin{pmatrix} 1 \\ 0 \end{pmatrix} & & \\ & \nearrow & & \searrow & \\ k & \xrightarrow{1} & k & \xrightarrow{\begin{pmatrix} 0 & 1 \end{pmatrix}} & k^2 \\ & & \downarrow \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} & \nearrow \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} & \\ & & k^3 & & \end{array}.$$

Indecomposable injectives.

The injective $I(1)$. Since 1 is a source, we have

$$I(1) = S(1): \begin{array}{ccccc} & & & & \\ & \nearrow & & \searrow & \\ k & \longrightarrow & 0 & \longrightarrow & 0 \\ & & \downarrow & \nearrow & \\ & & 0 & & \end{array}.$$

The injective $I(2)$. The paths ending at 2 are

$$e_2 \text{ (starting at 2),} \quad \alpha \text{ (starting at 1).}$$

Thus

$$I(2)_1 \cong k, \quad I(2)_2 \cong k, \quad I(2)_3 = 0, \quad I(2)_4 = 0,$$

and

$$I(2): \begin{array}{ccccc} & & 0 & & \\ & \nearrow & & \searrow & \\ k & \xrightarrow{1} & k & \xrightarrow{0} & 0 \\ & & \downarrow 0 & \nearrow & \\ & & 0 & & \end{array}.$$

The injective $I(3)$. The paths ending at 3 are

$$e_3 \text{ (starting at 3),} \quad \gamma \text{ (starting at 2),} \quad \beta, \gamma\alpha \text{ (starting at 1).}$$

Hence

$$I(3)_1 \cong k^2, \quad I(3)_2 \cong k, \quad I(3)_3 \cong k, \quad I(3)_4 = 0.$$

With respect to the ordered basis $\{\beta, \gamma\alpha\}$ at vertex 1, we obtain

$$I(3): \begin{array}{ccccc} & & (1,0) & & \\ & \nearrow & & \searrow & \\ k^2 & \xrightarrow{(0,1)} & k & \xrightarrow{1} & k \\ & \downarrow 0 & & \swarrow 0 & \\ & 0 & & & \end{array} .$$

The injective $I(4)$. The paths ending at 4 are

$$\begin{aligned} e_4 \text{ (starting at 4),} \quad \epsilon \text{ (starting at 3),} \\ \delta, \epsilon\gamma \text{ (starting at 2),} \quad \delta\alpha, \epsilon\beta, \epsilon\gamma\alpha \text{ (starting at 1).} \end{aligned}$$

Therefore

$$I(4)_1 \cong k^3, \quad I(4)_2 \cong k^2, \quad I(4)_3 \cong k, \quad I(4)_4 \cong k.$$

Choose the ordered bases

$$I(4)_1 : \{\delta\alpha, \epsilon\beta, \epsilon\gamma\alpha\}, \quad I(4)_2 : \{\delta, \epsilon\gamma\}.$$

Then

$$I(4): \begin{array}{ccccc} & & (0,1,0) & & \\ & \nearrow & & \searrow & \\ k^3 & \xrightarrow{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}} & k^2 & \xrightarrow{(0,1)} & k \\ & \downarrow (1,0) & & \swarrow 1 & \\ & k & & & \end{array} .$$

This determines all indecomposable projective and injective representations for Q_1 and Q_2 . $\square$

- 15.** Compute a projective resolution of each simple representation $S(i)$ for the quivers Q_1 and Q_2 from Problem 14.

Proof. We compute the (minimal) projective resolutions of the simple representations for the two quivers from Problem 14.

Recall that for an acyclic quiver Q , the category $\text{Rep } Q$ is hereditary. Hence every representation has projective dimension at most 1. In particular, for each simple representation $S(i)$, its minimal projective resolution is obtained from the projective cover

$$P(i) \twoheadrightarrow S(i),$$

whose kernel is $\text{rad } P(i)$.

Moreover, for a path algebra of an acyclic quiver, one has

$$\text{rad } P(i) \cong \bigoplus_{\alpha: i \rightarrow j} P(j),$$

where the direct sum runs over all arrows starting at i .

(1) **The quiver Q_1 .**

$$Q_1: \quad 1 \xrightarrow[\alpha]{\beta} 2 \xrightarrow[\gamma]{\delta} 3$$

Resolution of $S(1)$. Since 1 is a sink, $P(1) = S(1)$ is already projective. Hence the minimal projective resolution is

$$0 \longrightarrow P(1) \xrightarrow{\cong} S(1) \longrightarrow 0.$$

Resolution of $S(2)$. There are exactly two arrows starting at 2, namely $\alpha, \beta : 2 \rightarrow 1$. Therefore

$$\text{rad } P(2) \cong P(1) \oplus P(1).$$

Thus we obtain a short exact sequence

$$0 \longrightarrow P(1) \oplus P(1) \xrightarrow{f} P(2) \xrightarrow{\pi} S(2) \longrightarrow 0,$$

where π is the canonical projection onto the top of $P(2)$, and f is the morphism whose two components correspond to the two arrows α and β .

More explicitly, if we denote by

$$\iota_\alpha, \iota_\beta : P(1) \rightarrow P(2)$$

the morphisms sending the generator of $P(1)_1 \cong k$ to the basis vectors

$$\alpha, \beta \in P(2)_1 \cong k^2,$$

then

$$f = (\iota_\alpha, \iota_\beta).$$

Its image is precisely the radical of $P(2)$, so this is the minimal projective resolution of $S(2)$.

Resolution of $S(3)$. There are exactly two arrows starting at 3, namely $\gamma, \delta : 3 \rightarrow 2$. Hence

$$\text{rad } P(3) \cong P(2) \oplus P(2).$$

Therefore

$$0 \longrightarrow P(2) \oplus P(2) \xrightarrow{g} P(3) \xrightarrow{\rho} S(3) \longrightarrow 0$$

is exact, where ρ is the canonical projection onto the top of $P(3)$, and g is induced by the arrows γ and δ .

Explicitly, if

$$\iota_\gamma, \iota_\delta : P(2) \rightarrow P(3)$$

are the morphisms sending the generator of $P(2)_2 \cong k$ to

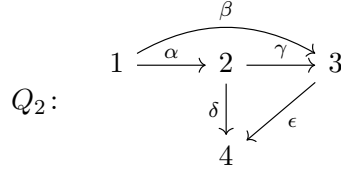
$$\gamma, \delta \in P(3)_2 \cong k^2,$$

then

$$g = (\iota_\gamma, \iota_\delta).$$

Its image is $\text{rad } P(3)$, so this gives the minimal projective resolution of $S(3)$.

(2) The quiver Q_2 .



Resolution of $S(4)$. Since 4 is a sink, $P(4) = S(4)$ is projective. Hence

$$0 \longrightarrow P(4) \xrightarrow{\cong} S(4) \longrightarrow 0$$

is the minimal projective resolution of $S(4)$.

Resolution of $S(3)$. There is exactly one arrow starting at 3, namely $\epsilon : 3 \rightarrow 4$. Thus

$$\text{rad } P(3) \cong P(4),$$

and we get

$$0 \longrightarrow P(4) \xrightarrow{\iota_\epsilon} P(3) \xrightarrow{\pi_3} S(3) \longrightarrow 0,$$

where ι_ϵ is the morphism sending the generator of $P(4)_4 \cong k$ to

$$\epsilon \in P(3)_4 \cong k.$$

This is the minimal projective resolution of $S(3)$.

Resolution of $S(2)$. There are two arrows starting at 2, namely

$$\gamma : 2 \rightarrow 3, \quad \delta : 2 \rightarrow 4.$$

Hence

$$\text{rad } P(2) \cong P(3) \oplus P(4).$$

Therefore

$$0 \longrightarrow P(3) \oplus P(4) \xrightarrow{h} P(2) \xrightarrow{\pi_2} S(2) \longrightarrow 0$$

is exact, where $h = (\iota_\gamma, \iota_\delta)$ and:

- $\iota_\gamma : P(3) \rightarrow P(2)$ sends the generator of $P(3)_3 \cong k$ to

$$\gamma \in P(2)_3 \cong k;$$

- $\iota_\delta : P(4) \rightarrow P(2)$ sends the generator of $P(4)_4 \cong k$ to

$$\delta \in P(2)_4 \cong k^2.$$

The image of h is exactly the radical of $P(2)$, so this is the minimal projective resolution of $S(2)$.

Resolution of $S(1)$. There are two arrows starting at 1, namely

$$\alpha : 1 \rightarrow 2, \quad \beta : 1 \rightarrow 3.$$

Thus

$$\text{rad } P(1) \cong P(2) \oplus P(3).$$

Hence we have the exact sequence

$$0 \longrightarrow P(2) \oplus P(3) \xrightarrow{u} P(1) \xrightarrow{\pi_1} S(1) \longrightarrow 0,$$

where $u = (\iota_\alpha, \iota_\beta)$ and:

- $\iota_\alpha : P(2) \rightarrow P(1)$ sends the generator of $P(2)_2 \cong k$ to

$$\alpha \in P(1)_2 \cong k;$$

- $\iota_\beta : P(3) \rightarrow P(1)$ sends the generator of $P(3)_3 \cong k$ to

$$\beta \in P(1)_3 \cong k^2.$$

Its image is $\text{rad } P(1)$, so this yields the minimal projective resolution of $S(1)$.

Summary.

For Q_1 :

$$\begin{aligned} 0 \rightarrow P(1) &\xrightarrow{\cong} S(1) \rightarrow 0, \\ 0 \rightarrow P(1)^2 &\rightarrow P(2) \rightarrow S(2) \rightarrow 0, \\ 0 \rightarrow P(2)^2 &\rightarrow P(3) \rightarrow S(3) \rightarrow 0. \end{aligned}$$

For Q_2 :

$$\begin{aligned} 0 \rightarrow P(4) &\xrightarrow{\cong} S(4) \rightarrow 0, \\ 0 \rightarrow P(4) &\rightarrow P(3) \rightarrow S(3) \rightarrow 0, \\ 0 \rightarrow P(3) \oplus P(4) &\rightarrow P(2) \rightarrow S(2) \rightarrow 0, \\ 0 \rightarrow P(2) \oplus P(3) &\rightarrow P(1) \rightarrow S(1) \rightarrow 0. \end{aligned}$$

These are precisely the minimal projective resolutions of the simple representations. $\square$

16. Show that for each $i \in Q_0$, the sequence

$$0 \longrightarrow \text{rad } P(i) \longrightarrow P(i) \longrightarrow S(i) \longrightarrow 0$$

is a minimal projective resolution.

Proof. Let Q be an acyclic quiver and let $i \in Q_0$. We must show that

$$0 \longrightarrow \text{rad } P(i) \xrightarrow{\iota} P(i) \xrightarrow{\pi} S(i) \longrightarrow 0$$

is a projective resolution that is minimal.

Step 1: The sequence is exact.

Recall that $P(i)_i \cong k$ with basis $\{e_i\}$, and for $j \neq i$, the space $P(i)_j$ has basis the set of paths from i to j . The simple representation $S(i)$ has $S(i)_i = k$ and $S(i)_j = 0$ for $j \neq i$.

Define $\pi: P(i) \rightarrow S(i)$ by

$$\pi_i = \text{id}_k: P(i)_i \rightarrow S(i)_i, \quad \pi_j = 0: P(i)_j \rightarrow 0 \text{ for } j \neq i.$$

This is clearly a surjective morphism of representations. Its kernel at each vertex is

$$\ker \pi_i = 0, \quad \ker \pi_j = P(i)_j \text{ for } j \neq i.$$

By definition, $\text{rad } P(i)$ has

$$(\text{rad } P(i))_i = 0, \quad (\text{rad } P(i))_j = P(i)_j \text{ for } j \neq i,$$

with the structure maps inherited from $P(i)$ except that arrows starting at i are set to zero. Since $\ker \pi$ as a subrepresentation of $P(i)$ has the same vector spaces, and the structure maps of $P(i)$ restricted to $\ker \pi$ automatically send 0 at vertex i to $P(i)_j$ via the zero map (for arrows α with $s(\alpha) = i$), we see that $\ker \pi = \text{rad } P(i)$. Hence the inclusion $\iota: \text{rad } P(i) \hookrightarrow P(i)$ gives the exact sequence

$$0 \longrightarrow \text{rad } P(i) \xrightarrow{\iota} P(i) \xrightarrow{\pi} S(i) \longrightarrow 0.$$

Step 2: $\text{rad } P(i)$ is projective.

If i is a sink, there are no arrows starting at i , so $P(i) = S(i)$ and $\text{rad } P(i) = 0$, which is trivially projective.

If i is not a sink, the radical decomposes as

$$\text{rad } P(i) \cong \bigoplus_{\substack{\alpha \in Q_1 \\ s(\alpha)=i}} P(t(\alpha)).$$

Indeed, each path from i to j (with $j \neq i$) begins with a unique first arrow α starting at i , and removing this first arrow gives a path from $t(\alpha)$ to j . This bijection on path

bases induces the isomorphism. Since each $P(t(\alpha))$ is projective and a direct sum of projectives is projective, $\text{rad } P(i)$ is projective.

Step 3: The resolution is minimal.

A projective resolution

$$0 \longrightarrow P_1 \xrightarrow{f} P_0 \xrightarrow{g} M \longrightarrow 0$$

is minimal if $g: P_0 \rightarrow M$ is a projective cover and $f: P_1 \rightarrow \ker g$ is a projective cover.

$\pi: P(i) \rightarrow S(i)$ is a projective cover. Let $g': P' \rightarrow S(i)$ be any surjective morphism with P' projective. Since $S(i)_i = k \neq 0$, there exists a nonzero element in P'_i mapping to $e_i \in S(i)_i$. By the isomorphism $\text{Hom}(P(i), P') \cong P'_i$, this element determines a morphism $h: P(i) \rightarrow P'$. The commutativity $g' \circ h = \pi$ follows from the universal property. Since π_i and g'_i are both surjections onto k , we can choose h so that h is surjective. Hence $P(i)$ with π is a projective cover of $S(i)$.

$\iota: \text{rad } P(i) \hookrightarrow P(i)$ restricted to $\ker \pi$ is a projective cover. The map ι identifies $\text{rad } P(i)$ with $\ker \pi$, and this is already an isomorphism $\text{rad } P(i) \xrightarrow{\sim} \ker \pi$. If $g'': P'' \twoheadrightarrow \ker \pi$ is any surjection from a projective P'' , then since $\text{rad } P(i)$ is projective (by Step 2), the identity on $\ker \pi$ lifts to a surjection $P'' \twoheadrightarrow \text{rad } P(i)$.

More directly, any proper direct summand of $\text{rad } P(i) \cong \bigoplus_{\alpha} P(t(\alpha))$ cannot surject onto $\text{rad } P(i)$, since the indecomposable projectives $P(t(\alpha))$ have $\text{End}(P(t(\alpha))) \cong k$ and thus satisfy the conditions for a projective cover.

Therefore

$$0 \longrightarrow \text{rad } P(i) \longrightarrow P(i) \longrightarrow S(i) \longrightarrow 0$$

is a minimal projective resolution. □

- 17.** Let $M = (M_i, \psi_{\alpha})$ be a representation of Q . Prove that for any vertex i in Q , there is an isomorphism of vector spaces:

$$\text{Hom}(M, I(i)) \cong M_i.$$

Proof. Recall that $I(i)_j$ has as a basis the set of all paths from j to i in Q , and that for an arrow $\alpha: j \rightarrow \ell$, the structure map

$$I(i)_{\alpha}: I(i)_j \rightarrow I(i)_{\ell}$$

is defined on a basis element (a path $p: j \rightarrow i$) by

$$I(i)_{\alpha}(p) = \begin{cases} p' & \text{if } p = p'\alpha, \text{ i.e., } p \text{ starts with } \alpha, \\ 0 & \text{otherwise.} \end{cases}$$

In particular, $I(i)_i \cong k$ with basis $\{e_i\}$ (the trivial path at i).

Define

$$\Phi: \text{Hom}(M, I(i)) \longrightarrow M_i^* = \text{Hom}_k(M_i, k)$$

by $\Phi(f) = f_i$, where we identify $I(i)_i \cong k$ via $e_i \mapsto 1$.

Φ is injective. Suppose $f: M \rightarrow I(i)$ is a morphism with $f_i = 0$. Let $j \in Q_0$ and let $m \in M_j$. We must show $f_j(m) = 0$, i.e., that for every path $p: j \rightarrow i$, the coefficient of p in $f_j(m)$ is zero. Write $p = \alpha_r \cdots \alpha_1$ where $\alpha_k: j_{k-1} \rightarrow j_k$ with $j_0 = j$ and $j_r = i$. Since f is a morphism, commutativity gives

$$f_i \circ \psi_{\alpha_r} \circ \cdots \circ \psi_{\alpha_1} = I(i)_{\alpha_r} \circ \cdots \circ I(i)_{\alpha_1} \circ f_j.$$

Under the identification $I(i)_i \cong k$, the composition $I(i)_{\alpha_r} \circ \cdots \circ I(i)_{\alpha_1}$ applied to $f_j(m)$ extracts exactly the coefficient of p in $f_j(m)$. Since $f_i = 0$, the left-hand side vanishes, so every such coefficient is zero. Hence $f_j = 0$ for all j , and Φ is injective.

Φ is surjective. Let $\lambda \in M_i^* = \text{Hom}_k(M_i, k)$. We construct a morphism $f^\lambda: M \rightarrow I(i)$ with $f_i^\lambda = \lambda$. For each vertex j and each $m \in M_j$, define

$$f_j^\lambda(m) = \sum_{p: j \rightarrow i} \lambda(\psi_p(m)) \cdot p,$$

where $\psi_p = \psi_{\alpha_r} \circ \cdots \circ \psi_{\alpha_1}$ denotes the composition of the structure maps of M along the path $p = \alpha_r \cdots \alpha_1$. This is a finite sum since Q is acyclic. In particular, $f_i^\lambda(m) = \lambda(m) \cdot e_i$, which corresponds to λ under the identification $I(i)_i \cong k$.

We verify that f^λ is a morphism. Let $\alpha: j \rightarrow \ell$ be an arrow. For $m \in M_j$:

$$f_\ell^\lambda(\psi_\alpha(m)) = \sum_{q: \ell \rightarrow i} \lambda(\psi_q(\psi_\alpha(m))) \cdot q = \sum_{q: \ell \rightarrow i} \lambda(\psi_{q\alpha}(m)) \cdot q.$$

On the other hand,

$$I(i)_\alpha(f_j^\lambda(m)) = \sum_{p: j \rightarrow i} \lambda(\psi_p(m)) \cdot I(i)_\alpha(p).$$

A path $p: j \rightarrow i$ satisfies $I(i)_\alpha(p) \neq 0$ if and only if p starts with α , i.e., $p = q\alpha$ for some path $q: \ell \rightarrow i$. In that case $I(i)_\alpha(q\alpha) = q$. Hence

$$I(i)_\alpha(f_j^\lambda(m)) = \sum_{q: \ell \rightarrow i} \lambda(\psi_{q\alpha}(m)) \cdot q = f_\ell^\lambda(\psi_\alpha(m)).$$

Thus f^λ is a morphism and $\Phi(f^\lambda) = \lambda$, so Φ is surjective.

Conclusion. Since all vector spaces are finite-dimensional, $M_i^* \cong M_i$ as vector spaces (non-canonically). Therefore

$$\text{Hom}(M, I(i)) \cong M_i^* \cong M_i$$

as k -vector spaces. □

18. Using Problem 17, show that the natural embedding

$$M \longrightarrow \bigoplus_{i \in Q_0} I(i)^{\dim(\text{soc } M)_i}$$

is an injective envelope of M . (I think here should be $\dim(\text{soc } M)_i$ but not $\dim M_i$ to make sure that the embedding is essential injective.)

Proof. Let

$$m_i := \dim_k(\text{soc } M)_i \quad \text{and} \quad E := \bigoplus_{i \in Q_0} I(i)^{m_i}.$$

Since $\text{soc } M$ is semisimple and the simple representations of Q are precisely the $S(i)$, we have

$$\text{soc } M \cong \bigoplus_{i \in Q_0} S(i)^{m_i}.$$

By Problem 17, applied to the simple representation $S(j)$, we get

$$\text{Hom}(S(j), I(i)) \cong S(j)_i = \begin{cases} k & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

Thus for each i there is, up to scalar, a unique nonzero morphism

$$S(i) \longrightarrow I(i),$$

and there is no nonzero morphism $S(j) \rightarrow I(i)$ for $j \neq i$. Therefore the above decomposition of $\text{soc } M$ yields an injective morphism

$$u: \text{soc } M \hookrightarrow E$$

whose image is exactly $\text{soc } E$.

Now let

$$s: \text{soc } M \hookrightarrow M$$

be the inclusion. Since E is injective (being a direct sum of injective representations), the morphism u extends along s to a morphism

$$\iota: M \longrightarrow E$$

such that

$$\iota \circ s = u.$$

We claim that ι is injective. Suppose $\ker \iota \neq 0$. Since $\ker \iota$ is a nonzero finite-dimensional representation, it contains a simple subrepresentation. Let $S \subseteq \ker \iota$ be simple. Then $S \subseteq M$ is a simple subrepresentation, hence

$$S \subseteq \text{soc } M.$$

But $\iota|_{\text{soc } M} = u$ is injective, so ι cannot vanish on the nonzero subrepresentation S , a contradiction. Hence

$$\ker \iota = 0,$$

so ι is a monomorphism.

Next we show that $\iota(M)$ is an essential subrepresentation of E . Let $0 \neq X \subseteq E$ be any nonzero subrepresentation. Again, X contains a simple subrepresentation S . Since S is simple and $S \subseteq E$, we have

$$S \subseteq \text{soc } E = \text{im}(u) \subseteq \iota(M).$$

Therefore

$$0 \neq S \subseteq X \cap \iota(M),$$

so $X \cap \iota(M) \neq 0$. Thus $\iota(M)$ is essential in E .

Finally, let

$$j: M \hookrightarrow J$$

be any injective morphism into an injective representation J . Since J is injective and $\iota: M \hookrightarrow E$ is a monomorphism, there exists a morphism

$$h: E \longrightarrow J$$

such that

$$h \circ \iota = j.$$

We show that h is injective. If $\ker h \neq 0$, then since $\iota(M)$ is essential in E , we must have

$$\ker h \cap \iota(M) \neq 0.$$

But on $\iota(M)$ we have $h \circ \iota = j$, and j is injective, so $h|_{\iota(M)}$ is injective. Hence

$$\ker h \cap \iota(M) = 0,$$

a contradiction. Therefore $\ker h = 0$, so h is injective.

Thus for every injective morphism $j: M \hookrightarrow J$ into an injective representation J , there exists an injective morphism $h: E \hookrightarrow J$ such that

$$h \circ \iota = j.$$

Therefore $\iota: M \hookrightarrow E$ is an injective envelope of M .

Hence the natural embedding

$$M \longrightarrow \bigoplus_{i \in Q_0} I(i)^{\dim(\text{soc } M)_i}$$

is an injective envelope of M . □

19. Let i and j be vertices in the quiver Q . Prove that the vector space

$$\text{Hom}(I(i), I(j))$$

has a basis consisting of all paths from j to i in Q . In particular,

$$\text{End}(I(i)) \cong k.$$

Proof. By Problem 17, for any representation X and any vertex j , the isomorphism

$$\text{Hom}(X, I(j)) \cong X_j^*$$

holds (where $X_j^* = \text{Hom}_k(X_j, k) \cong X_j$ as a vector space in finite dimensions). Applying this with $X = I(i)$, we obtain

$$\text{Hom}(I(i), I(j)) \cong I(i)_j^*.$$

By definition, $I(i)_j$ is the k -vector space with basis the set of all paths from j to i in Q . Hence

$$\dim \text{Hom}(I(i), I(j)) = \dim I(i)_j = |\{\text{paths from } j \text{ to } i\}|.$$

We now exhibit an explicit basis. For each path $p: j \rightarrow i$, we construct a morphism $\theta^p: I(i) \rightarrow I(j)$. By the isomorphism $\text{Hom}(I(i), I(j)) \cong I(i)_j^*$, a morphism $f: I(i) \rightarrow I(j)$ corresponds to a linear functional on $I(i)_j$. The path $p \in I(i)_j$ is a basis element, and we let $p^* \in I(i)_j^*$ be the corresponding dual basis element (the functional satisfying $p^*(p) = 1$ and $p^*(q) = 0$ for any other basis path $q \neq p$ from j to i). Let θ^p be the morphism corresponding to p^* .

Explicitly, for a vertex ℓ and a basis element $c \in I(i)_\ell$ (a path $c: \ell \rightarrow i$), the morphism θ^p is given by

$$\theta_\ell^p(c) = \begin{cases} c' & \text{if } c = pc', \text{ i.e., the path } c \text{ factors as } \ell \xrightarrow{c'} j \xrightarrow{p} i, \\ 0 & \text{otherwise.} \end{cases}$$

Here, $c': \ell \rightarrow j$ is the remaining path after removing the suffix p . We verify this is a morphism. Let $\alpha: \ell \rightarrow \ell'$ be an arrow. For a basis element $c: \ell \rightarrow i$ of $I(i)_\ell$:

- The map $I(i)_\alpha(c)$ is nonzero only if c starts with α , say $c = c''\alpha$. In that case $I(i)_\alpha(c) = c'' \in I(i)_{\ell'}$.
- We need $\theta_{\ell'}^p(I(i)_\alpha(c)) = I(j)_\alpha(\theta_\ell^p(c))$.

If $c = pc'$ for some path $c': \ell \rightarrow j$, and also c starts with α (so c' starts with α , say $c' = c''\alpha$), then $I(i)_\alpha(c) = pc''$ and $\theta_{\ell'}^p(pc'') = c''$. On the other hand, $\theta_\ell^p(c) = c'$ and $I(j)_\alpha(c') = c''$ (since $c' = c''\alpha$ starts with α). Both sides agree. The other cases are similarly verified.

Since the paths p from j to i form a basis of $I(i)_j$, the dual functionals $\{p^*\}$ form a basis of $I(i)_j^*$, and thus the morphisms $\{\theta^p\}$ form a basis of $\text{Hom}(I(i), I(j))$.

The special case $i = j$. The only path from i to i is the trivial path e_i (since Q has no oriented cycles). Therefore

$$\text{End}(I(i)) = \text{Hom}(I(i), I(i)) \cong k,$$

with basis $\{\theta^{e_i}\} = \{\text{id}_{I(i)}\}$. □

20. Prove that the representation $I(j)$ is a simple representation if and only if

$$\text{Hom}(I(i), I(j)) = 0 \quad \text{for all } i \neq j.$$

Proof. By Problem 19, for any vertices $i, j \in Q_0$,

$$\text{Hom}(I(i), I(j)) \cong I(i)_j^*,$$

where $I(i)_j$ has basis the set of paths from j to i in Q .

($\Rightarrow$) Assume $I(j)$ is simple. Let $i \neq j$ and let $f: I(i) \rightarrow I(j)$ be a morphism. The image $\text{im}(f)$ is a subrepresentation of $I(j)$. Since $I(j)$ is simple, either $\text{im}(f) = 0$ or $\text{im}(f) = I(j)$.

Suppose $\text{im}(f) = I(j)$, so f is surjective. The exact sequence

$$0 \rightarrow \ker f \rightarrow I(i) \xrightarrow{f} I(j) \rightarrow 0$$

splits, because $I(j)$ is injective. Hence $I(i) \cong \ker f \oplus I(j)$. Since $I(i)$ is indecomposable and $I(j) \neq 0$, we get $\ker f = 0$ and $I(i) \cong I(j)$. But an isomorphism forces identical dimension vectors: in particular, $\dim I(i)_j = \dim I(j)_j = 1$, so there exists a path from j to i , and $\dim I(j)_i = \dim I(i)_i = 1$, so there exists a path from i to j . Composing these two nontrivial paths (since $i \neq j$) yields a nontrivial oriented cycle, contradicting the acyclicity of Q .

Therefore $\text{im}(f) = 0$, i.e., $f = 0$. Hence $\text{Hom}(I(i), I(j)) = 0$ for all $i \neq j$.

($\Leftarrow$) Assume $\text{Hom}(I(i), I(j)) = 0$ for all $i \neq j$. We show $I(j)$ is simple.

Suppose for contradiction that $I(j)$ is not simple, so there exists a nonzero proper subrepresentation $N \subsetneq I(j)$.

Since Q is acyclic, the path algebra kQ is hereditary, and thus $\text{Rep } Q$ is a hereditary category. In a hereditary category, subrepresentations of injective representations are injective. Therefore N is injective.

By the Krull–Schmidt theorem, $N \cong \bigoplus_{i \in Q_0} I(i)^{m_i}$ for some non-negative integers m_i , not all zero. The inclusion $N \hookrightarrow I(j)$ restricts to a nonzero morphism $I(i) \rightarrow I(j)$ for each i with $m_i > 0$.

If $i \neq j$: by hypothesis, $\text{Hom}(I(i), I(j)) = 0$, so no such nonzero morphism exists. Hence $m_i = 0$ for all $i \neq j$.

If $i = j$: then $N \cong I(j)^m$ for some $m \geq 1$. Since N is a subrepresentation of $I(j)$ and $I(j)$ is indecomposable, comparing dimensions gives

$$\dim N_\ell = m \cdot \dim I(j)_\ell \leq \dim I(j)_\ell$$

for every vertex ℓ . Since $I(j)_j \cong k \neq 0$, this forces $m \leq 1$, so $m = 1$ and $N \cong I(j)$, hence $N = I(j)$. This contradicts $N \subsetneq I(j)$.

Therefore no such N exists, and $I(j)$ is simple. $\square$

21. Show that for any vertex $i \in Q_0$,

$$D(P_Q(i)) = I_{Q^{\text{op}}}(i).$$

Proof. For each vertex $j \in Q_0$, let $\mathcal{P}_Q(i, j)$ denote the set of all paths from i to j in Q . By definition,

$$P_Q(i)_j = k\langle \mathcal{P}_Q(i, j) \rangle.$$

Therefore

$$D(P_Q(i))_j = \text{Hom}_k(P_Q(i)_j, k)$$

has dual basis

$$\{\varepsilon_c \mid c \in \mathcal{P}_Q(i, j)\}, \quad \varepsilon_c(c') = \delta_{c, c'}.$$

On the other hand, reversing arrows gives a bijection

$$\mathcal{P}_Q(i, j) \longrightarrow \mathcal{P}_{Q^{\text{op}}}(j, i), \quad c \longmapsto c^{\text{op}}.$$

Thus, by the definition of the injective representation over Q^{op} ,

$$I_{Q^{\text{op}}}(i)_j = k\langle \mathcal{P}_{Q^{\text{op}}}(j, i) \rangle$$

has basis

$$\{c^{\text{op}} \mid c \in \mathcal{P}_Q(i, j)\}.$$

So for each $j \in Q_0$ we obtain a vector space isomorphism

$$\Phi_j: D(P_Q(i))_j \longrightarrow I_{Q^{\text{op}}}(i)_j, \quad \Phi_j(\varepsilon_c) = c^{\text{op}}.$$

It remains to check that these maps are compatible with the arrow maps. Let $\alpha: j \rightarrow \ell$ be an arrow in Q . Then $\alpha^{\text{op}}: \ell \rightarrow j$ is an arrow in Q^{op} .

In $P_Q(i)$, the map

$$\varphi_\alpha: P_Q(i)_j \rightarrow P_Q(i)_\ell$$

is given on basis paths by

$$\varphi_\alpha(c) = c\alpha.$$

Hence in $D(P_Q(i))$, the map corresponding to α^{op} is

$$D(\varphi_\alpha): D(P_Q(i)_\ell) \rightarrow D(P_Q(i)_j), \quad f \mapsto f \circ \varphi_\alpha.$$

Now let $p \in \mathcal{P}_Q(i, \ell)$. For any basis path $c \in \mathcal{P}_Q(i, j)$, we have

$$D(\varphi_\alpha)(\varepsilon_p)(c) = \varepsilon_p(\varphi_\alpha(c)) = \varepsilon_p(c\alpha) = \begin{cases} 1, & p = c\alpha, \\ 0, & \text{otherwise.} \end{cases}$$

Therefore

$$D(\varphi_\alpha)(\varepsilon_p) = \begin{cases} \varepsilon_c, & p = c\alpha, \\ 0, & \text{otherwise.} \end{cases}$$

Under the bijection of paths, the condition $p = c\alpha$ is equivalent to

$$p^{\text{op}} = \alpha^{\text{op}} c^{\text{op}}.$$

But this is exactly the defining rule for the arrow map

$$\psi_{\alpha^{\text{op}}}: I_{Q^{\text{op}}}(i)_\ell \rightarrow I_{Q^{\text{op}}}(i)_j,$$

namely,

$$\psi_{\alpha^{\text{op}}}(p^{\text{op}}) = \begin{cases} c^{\text{op}}, & p^{\text{op}} = \alpha^{\text{op}} c^{\text{op}}, \\ 0, & \text{otherwise.} \end{cases}$$

Thus

$$\Phi_j \circ D(\varphi_\alpha) = \psi_{\alpha^{\text{op}}} \circ \Phi_\ell.$$

So the family $(\Phi_j)_{j \in Q_0}$ is an isomorphism of representations

$$D(P_Q(i)) \cong I_{Q^{\text{op}}}(i).$$

This proves the claim. □